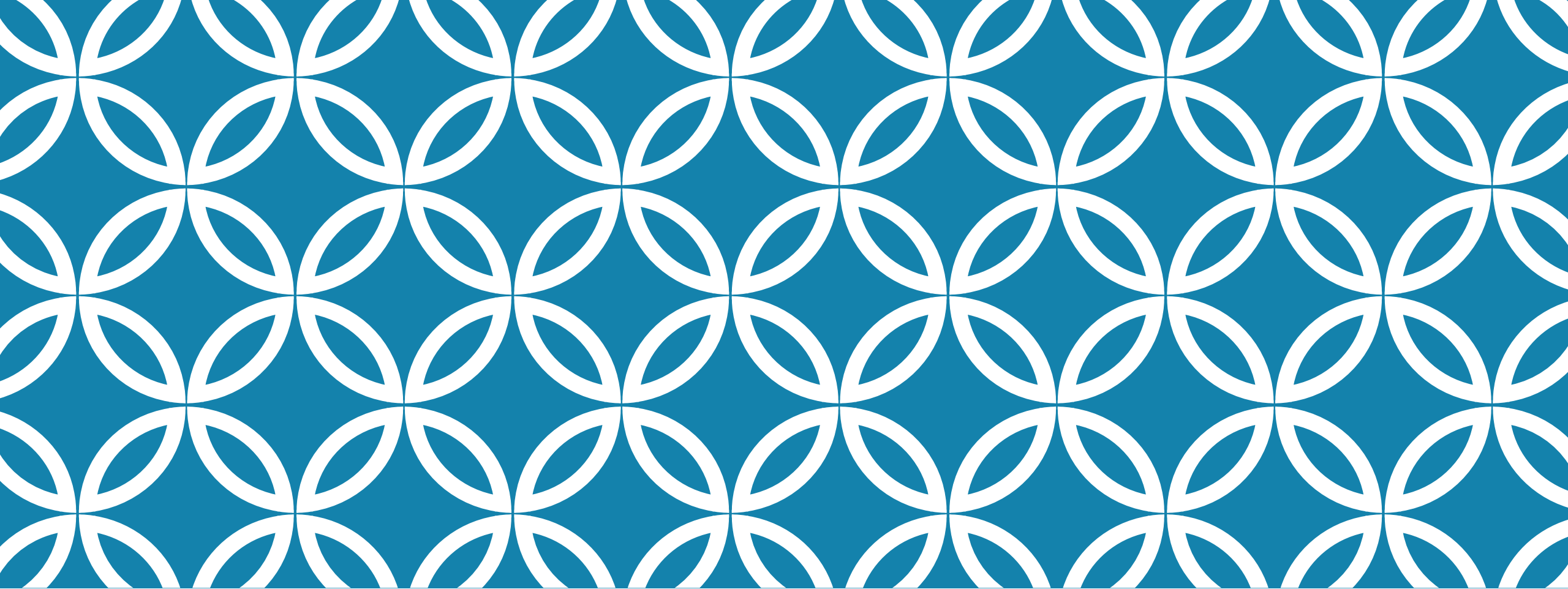




METAGENOMICS WORKSHOP

Avril Metcalfe-Roach
4 Dec 2024

TOPICS COVERED TODAY

1. **Metagenomic sequencing principles**

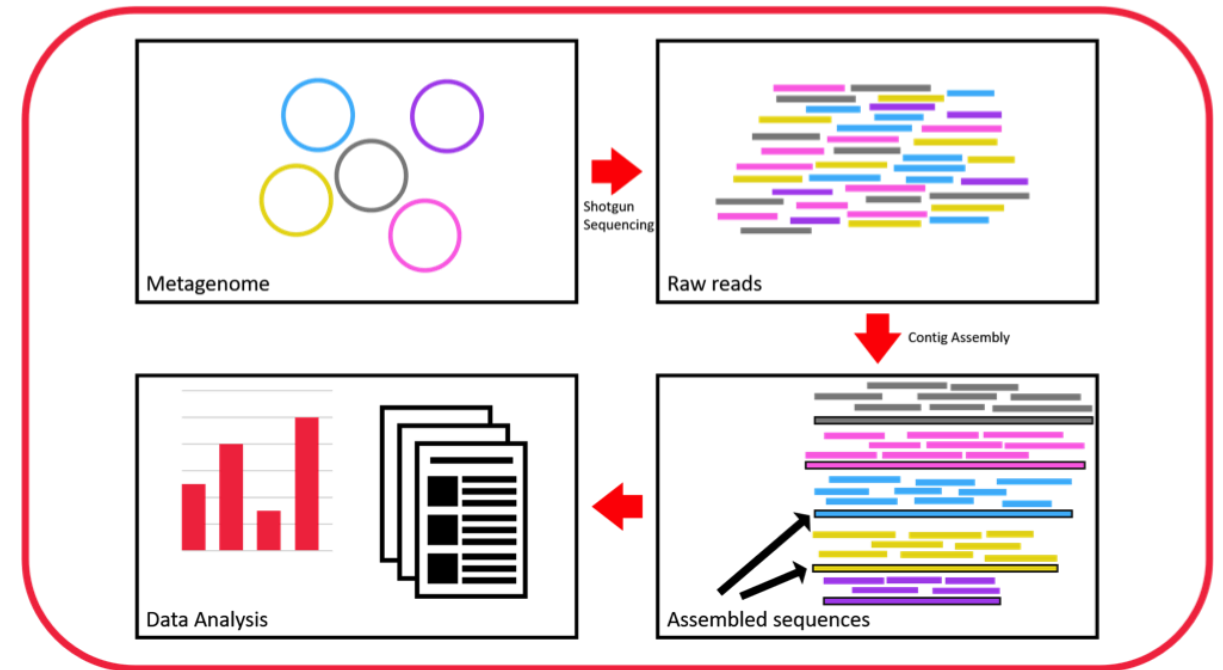
1. How sequencing works
2. Metagenomics pipeline
3. Sequencing data distribution vs. the actual microbiome
4. Data normalization
5. Compositional Data Analysis: Log Transformations
6. Overabundance of Zeros

2. Raw Data Processing (Tutorial)

3. Key R Analyses (Tutorial)

HOW SEQUENCING WORKS: PRINCIPLE

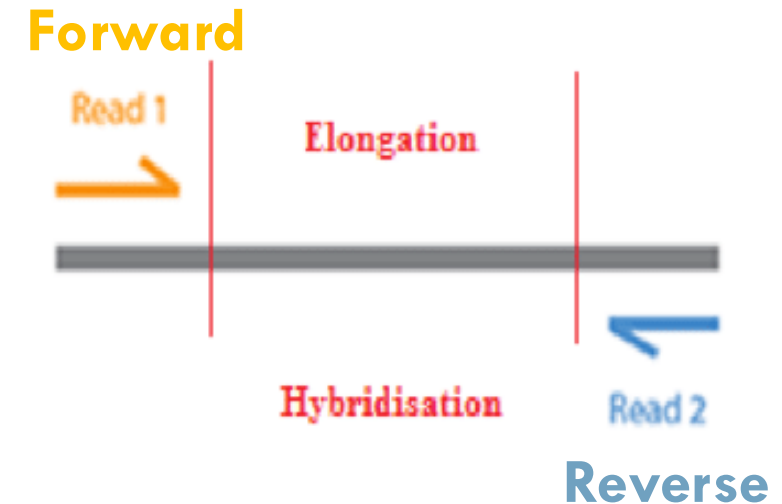
- **Shotgun metagenomic sequencing:**
DNA is randomly fragmented,
fragments are sequenced
- Sequences represent ENTIRE
metagenome, not just taxonomic
markers (16S sequencing)
 - Directly measure **functional
capacity**



Bio Basic Asia Pacific

HOW SEQUENCING WORKS: DATA QUANTITY

- Each read is a **standard length**
 - Usually 100 or 150 bp per read
- **Paired-end sequencing:** each read is read **forward and back** (typical for DNA)
 - More expensive, but better quality data
 - 2x number of files (R1, R2)
- Example sequencing parameters: “2x150 bp with an average depth of 10 million reads/sample”
 - Paired end, 150 bp length, 10 million reads PER DIRECTION*



Study Accession	Scientific Name	Download All	
		Generated FASTQ files: FTP	
PRJNA834801	human gut metagenome	☐	SRR19064318_1.fastq.gz
		☐	SRR19064318_2.fastq.gz

*Note: some facilities provide sequencing quotes based on total reads (both directions).

HOW SEQUENCING WORKS: DATA QUANTITY

Sequencing depth:
of reads



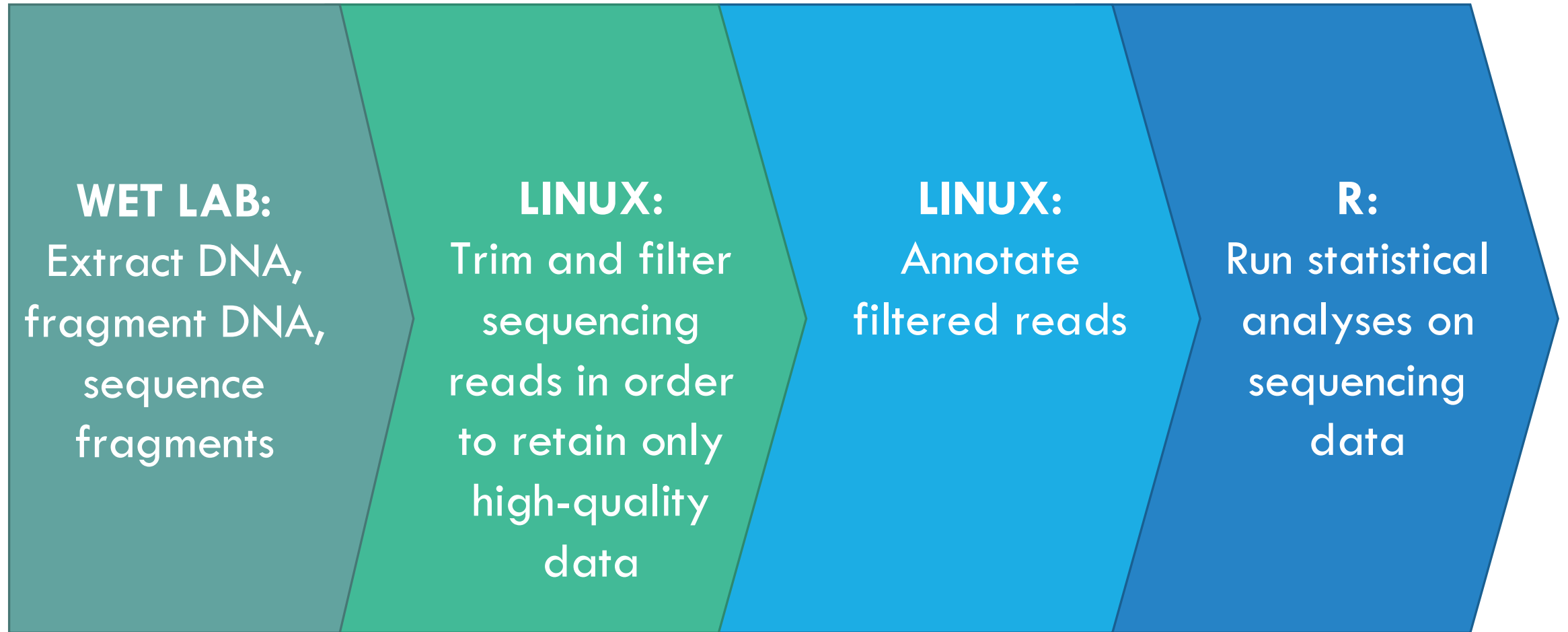
CATGCTACGTCGTATGCTGACTGATGCTGATGCTGCATGCATGCAT
TACCTAGCTGAGCATCGTATCGATCGTACGTACGTAGTCGTCTGAA
ATCTCGATAACGACTACGATCACTACGCTAGCTAGCACGTACGTAC

Read length: # nt per read

Total # bases: Sequencing depth * Read Length (*2 if paired end)

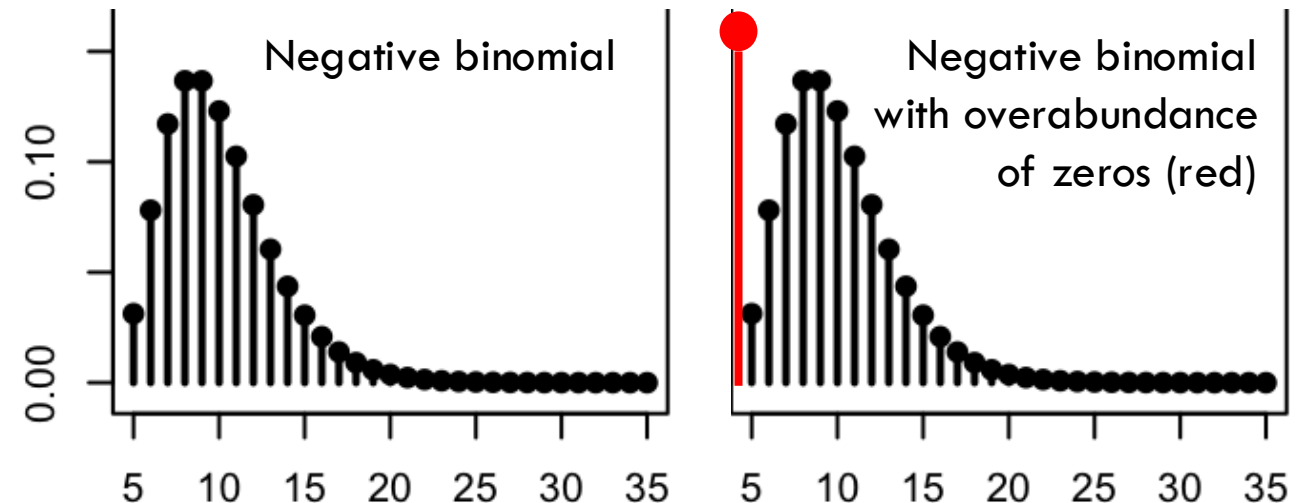
Gigabase/Giganucleotide (Gb/Gnt): 1 billion bases (10^9)

METAGENOMICS PIPELINE

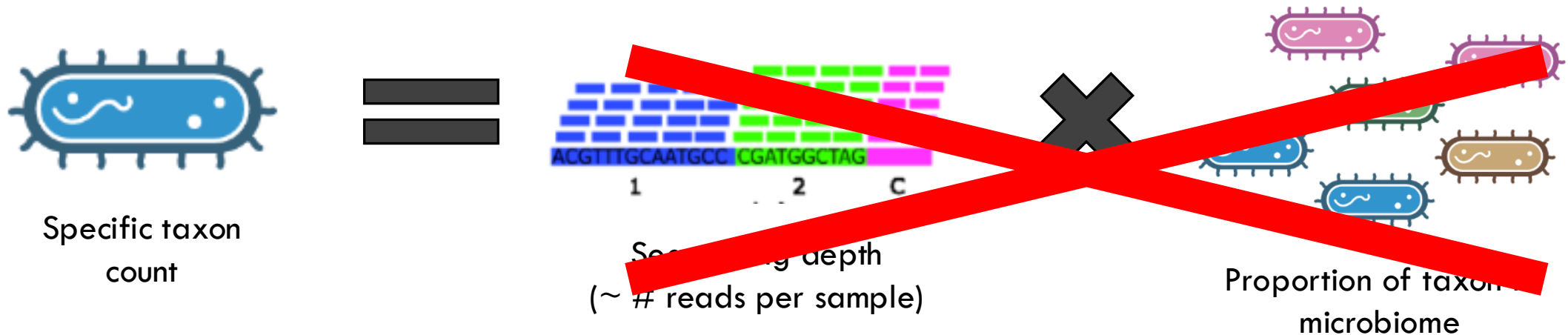


MICROBIOME DATA IS NOT NORMAL.

- Often described as a **negative binomial distribution**
 - No true higher bound, but can't have fewer than zero microbes
- HOWEVER, microbiome data also has an **overabundance of zero values** – not accounted for by traditional normalization
 - Ex. Half the population follows a negative binomial distribution, other half have zero abundance
- There is **no perfect way to normalize** most microbiome data
 - Most tools have in-built methods

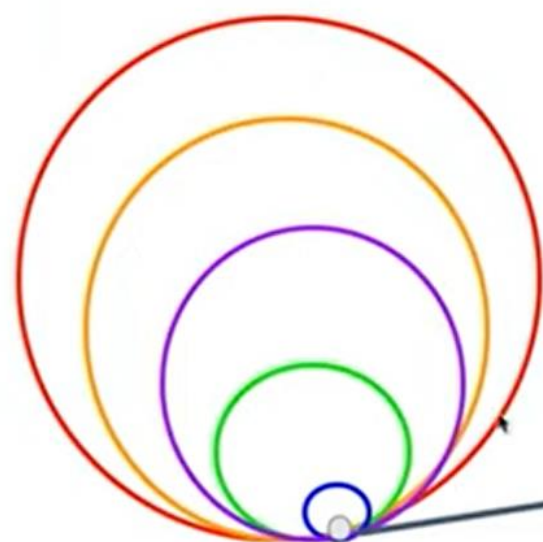


MICROBIOME SEQUENCING DATA IS COMPOSITIONAL.



- *Compositional = relative abundance (percentages)*
- Proportion of taxa: relatively high consistency between runs
- Sequencing depth: relatively **prone to technical variation**
 - Ex. PCR bias (specific taxa, kits, settings), variance in sample concentration, random fluctuations

Steps from sampling to sequencing and analysis add bias to observed composition



- Total bacterial population
- Sampled portion
- DNA/RNA sample
- Amplified product
- Library for sequencing
- Reads from the sequencer

A

Actual composition

● : 33%
● : 33%
● : 33%



Extraction



PCR



Sequencing



Bioinformatics

Observed composition

● : 4%
● : 72%
● : 24%

McLaren et al. 2019 "Consistent and correctable bias in metagenomic sequencing experiments"

www.dnagenotek.com

ACTUAL MICROBIOMES ARE COMPOSITIONAL AND ABSOLUTE

Some taxa *may* impact the abundance of other taxa

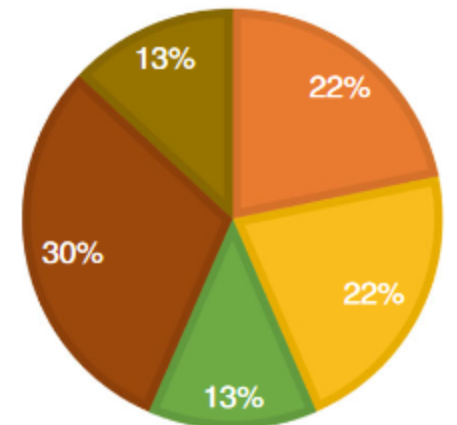
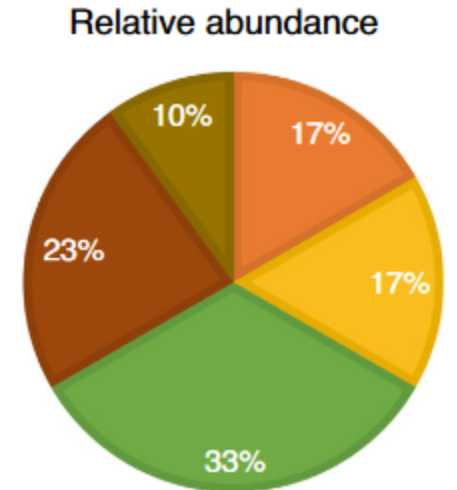
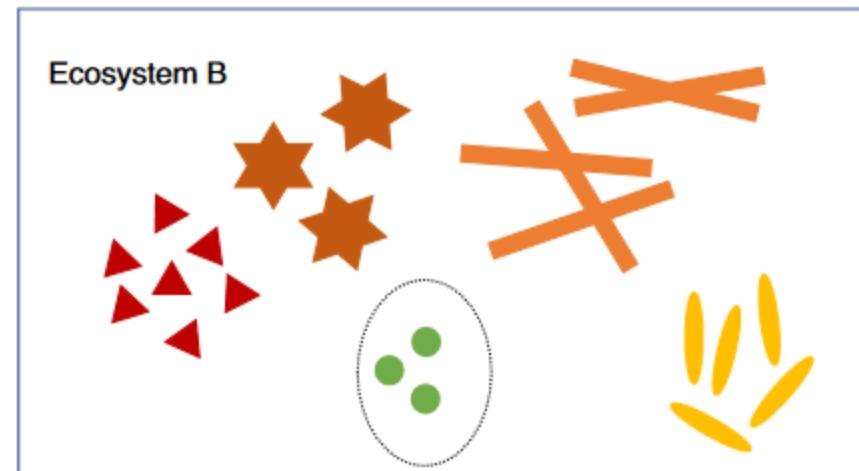
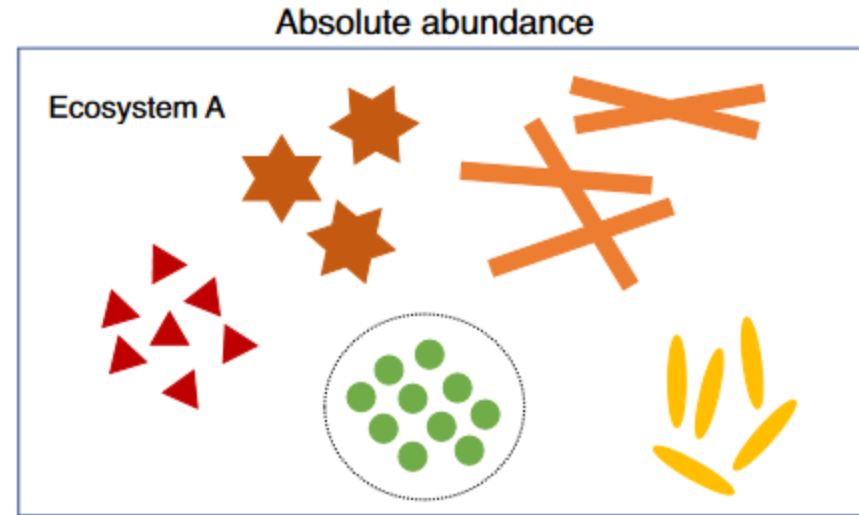
- Competing resources, crossfeeding, etc

Data cannot easily account for differences in **absolute abundance**

- Absolute increase in 1 taxon = **relative** decrease in **all other taxa**

Difficult to accurately measure absolute abundance per sequencing sample

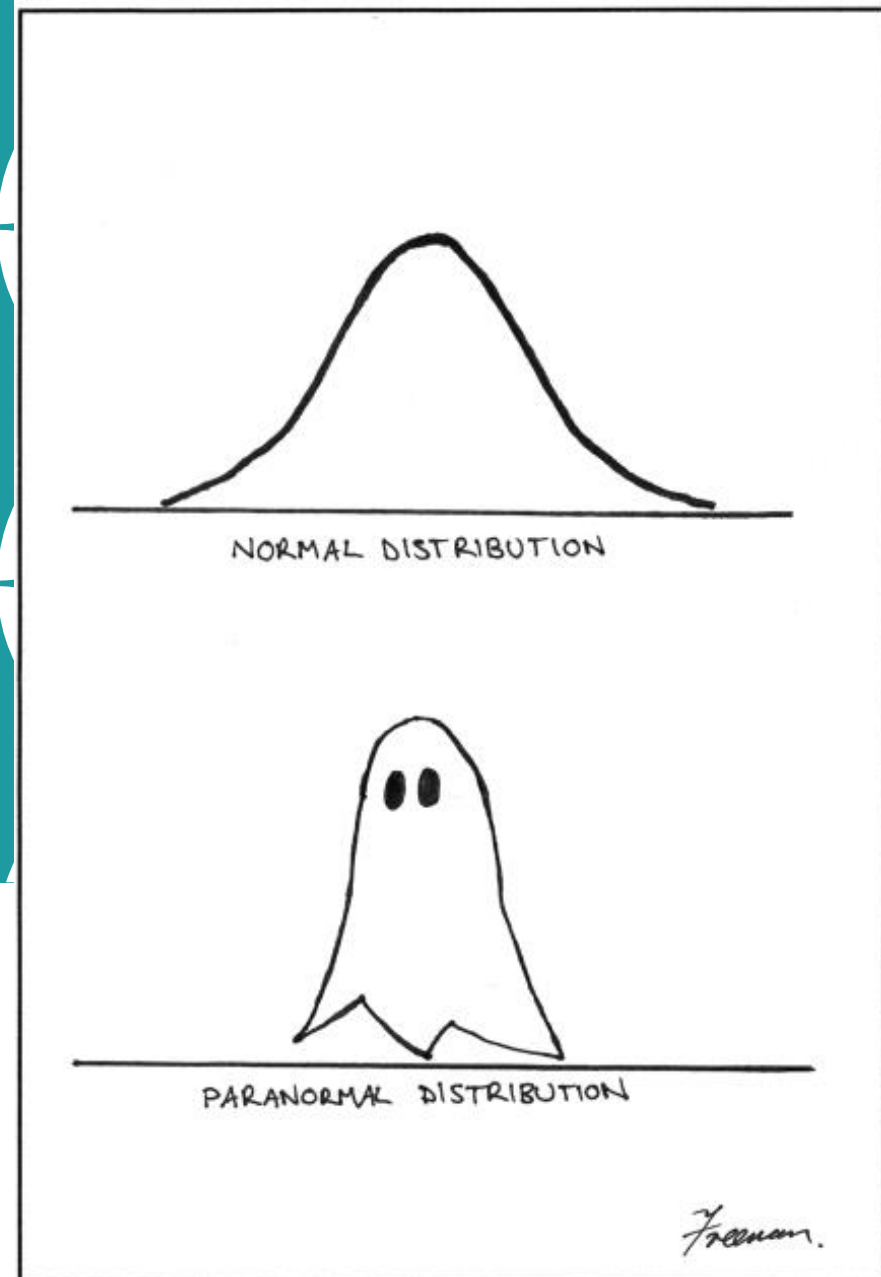
- Too much artefactual error



Statistically, we don't have a way to perfectly address these foundational issues. *(Though people are trying!)*

The best we can do is be aware of the limitations, address what we can, and keep the rest in mind when analyzing the data.

NORMALIZATION AND TRANSFORMATION METHODS

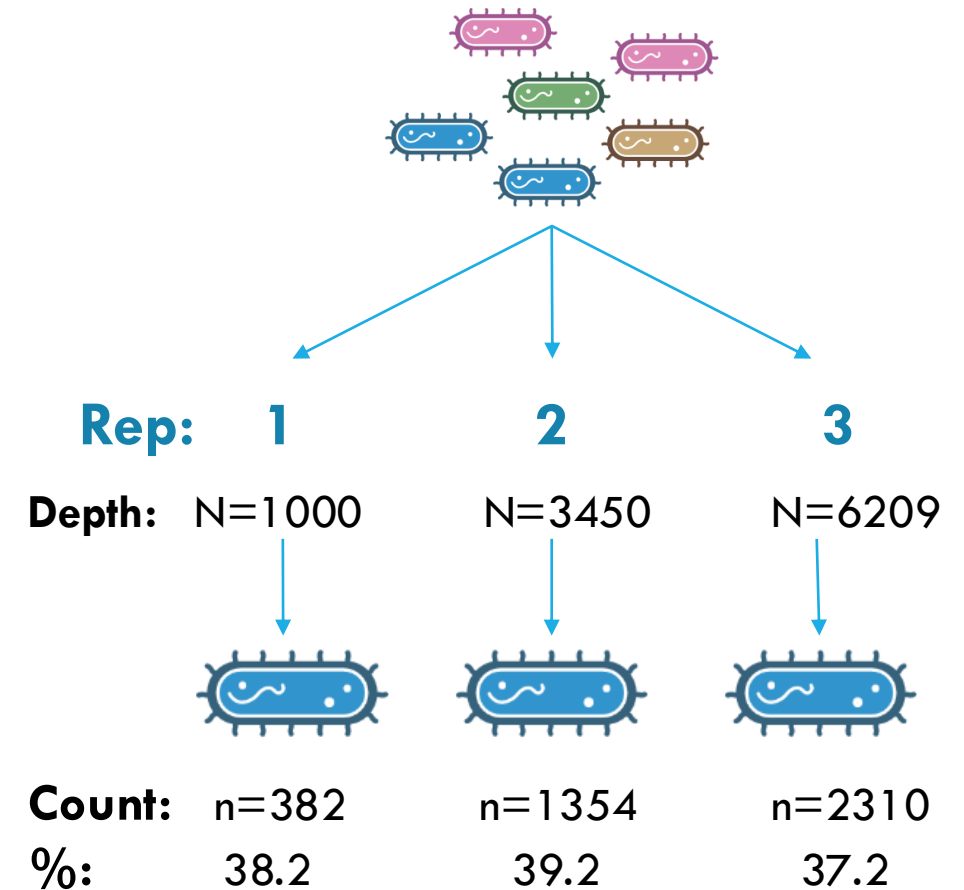


MOST COMMON TECHNIQUES

Rarefaction: randomly subsample reads to match smallest sample depth

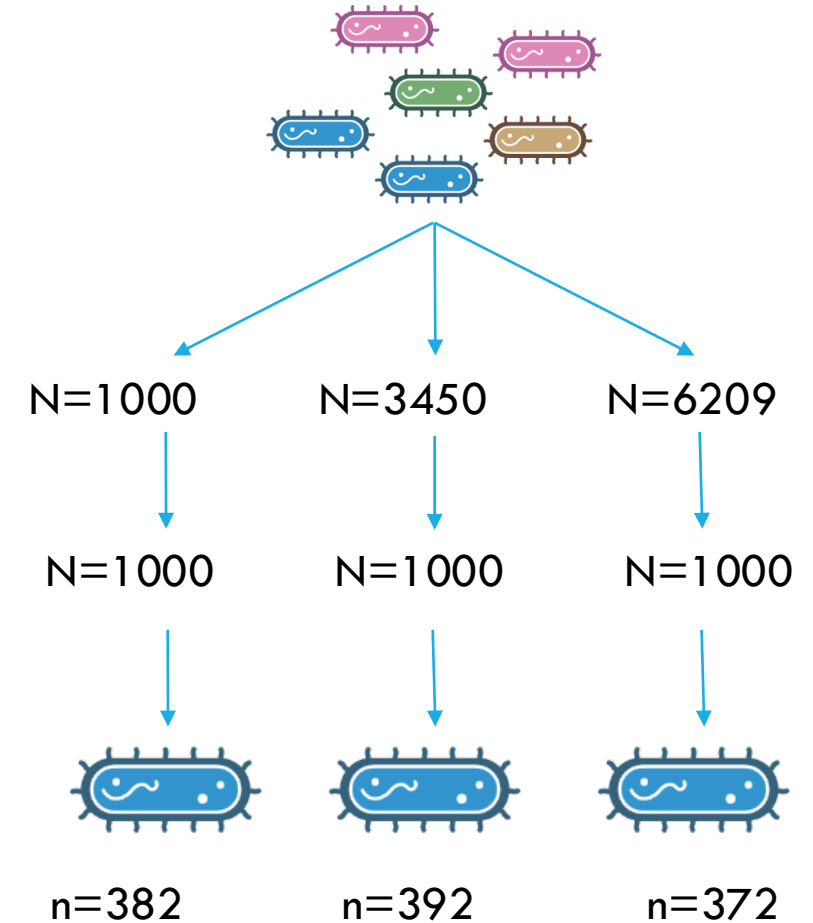
Relative abundance: divide by total sample reads

Log or Center-Log-Ratio (CLR) transformations: addresses the negative binomial distribution



RAREFACTION: *RARE*-LY USED

- Randomly selects reads up to **pre-defined minimum**
 - Minimum: typically smallest usable sample
- Primarily used to calculate **diversity metrics**
 - Diversity: richness, evenness, similarity
 - Larger depth = more rare taxa detected, can **artefactually increase diversity**
- Occasionally recommended by other tools
- **Problematic for many analyses, such as differential abundance:**
 - **Exclusion of valid data** - reduced statistical power
 - Chosen read depth is arbitrary/inconsistent

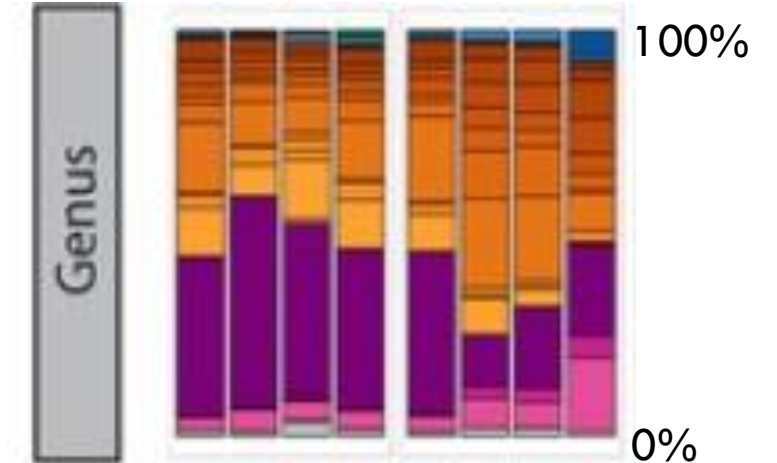


RELATIVE ABUNDANCE

AKA compositional, % abundance, total sum scaled (TSS)

Most intuitive & very common: gives bacterial proportions (sum to 100%)

Used **extensively** for normalization AND data visualization



Raw Counts

Taxon	Person 1	Person 2
Taxon 1	100	300
Taxon 2	54	154
Total Reads	154	454

Relative Abundance

Taxon	Person 1	Person 2
Taxon 1	0.65	0.66
Taxon 2	0.35	0.34
Total Reads	1	1

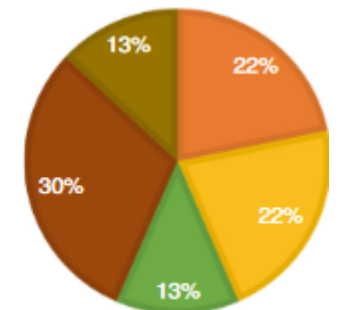
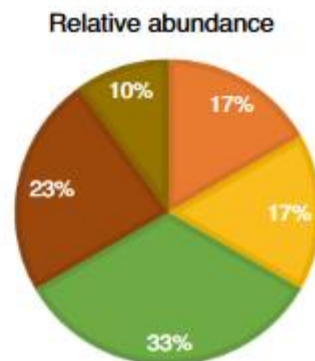
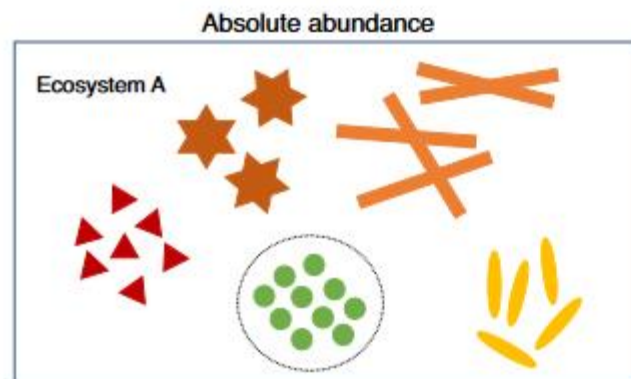
PITFALLS OF RELATIVE ABUNDANCE

Problem 1: it produces data whose values are **inter-dependent**.

Change in 1 taxon = change in ALL taxa

Traditional statistical methods assume values are **independent**

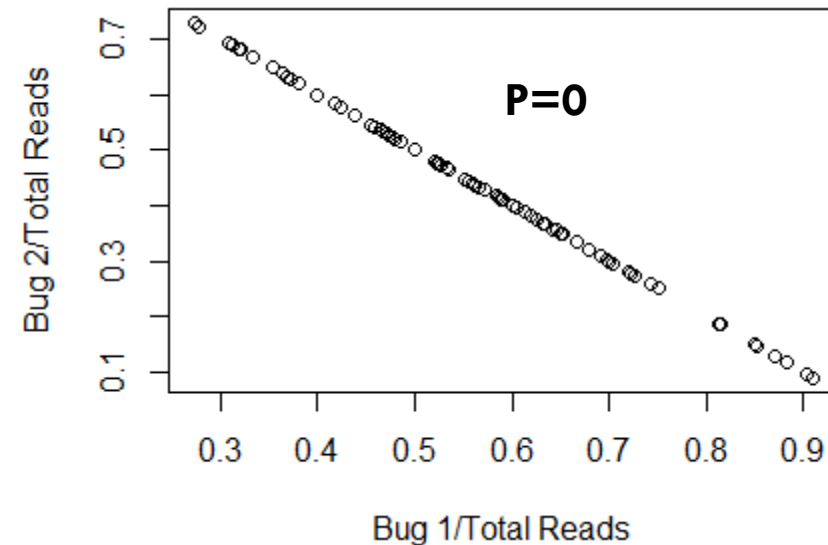
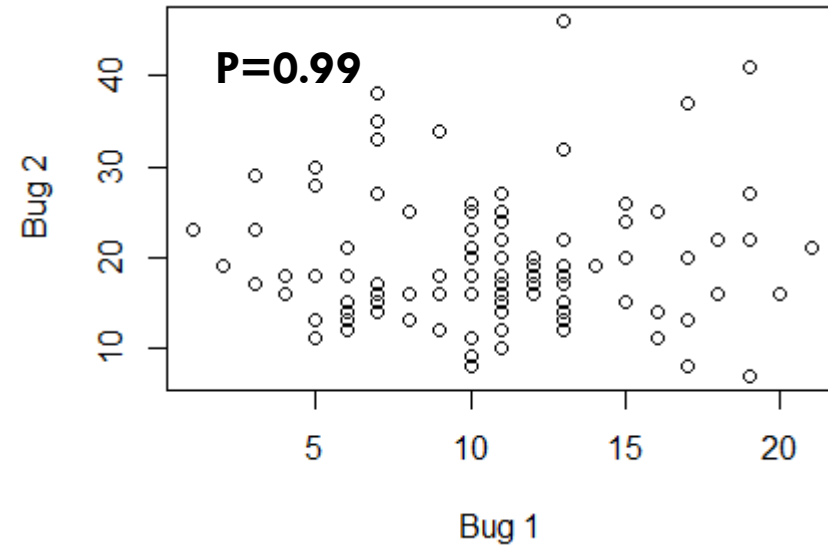
- Ex. linear models, correlation coefficients, PCAs
- **Overestimate strength of correlations**
- CAN still use these statistical methods if certain transformations are applied (**log-ratio transforms**)



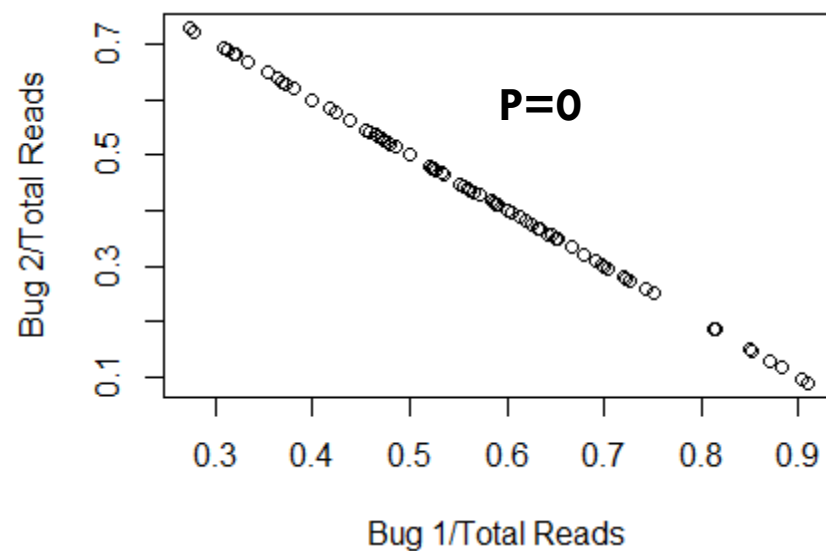
VISUALIZATION OF RELATIVE ABUNDANCE BIAS

1. Generate two **random** distributions:
Bug 1, Bug 2
 - Check that they don't correlate by chance
2. Calculate totals per sample (Bug 1 + Bug 2)
3. Calculate relative abundance for each bug
4. Correlate Bug 1 & 2

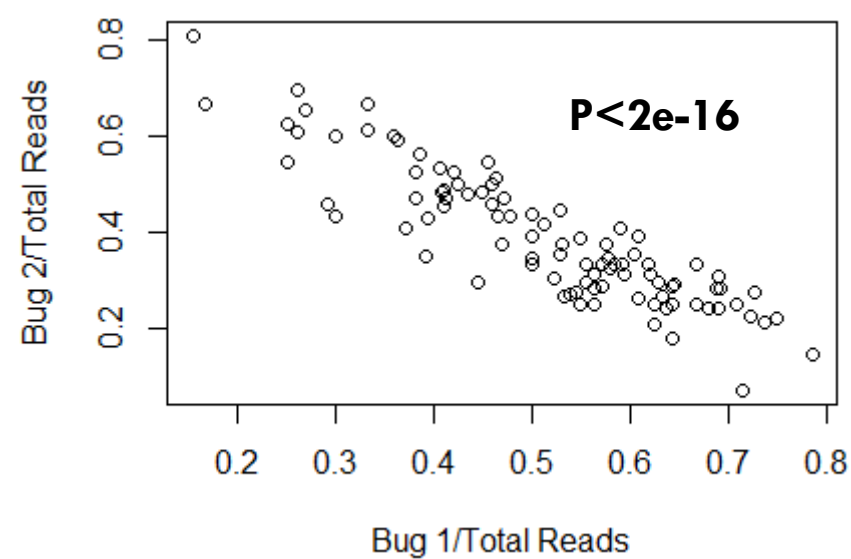
Because the total must equal 1, 2 bugs will always be perfectly correlated. 3% must be matched with 97%, 5% with 95%, etc.



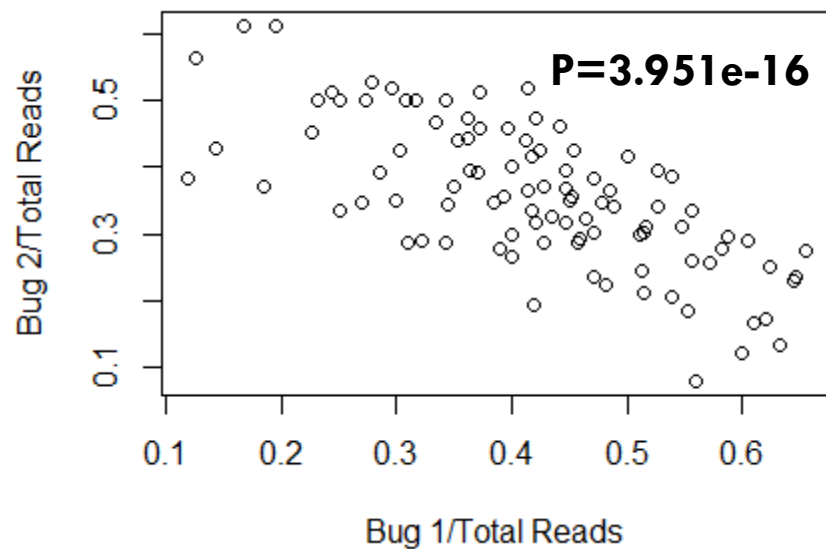
N = 2 taxa



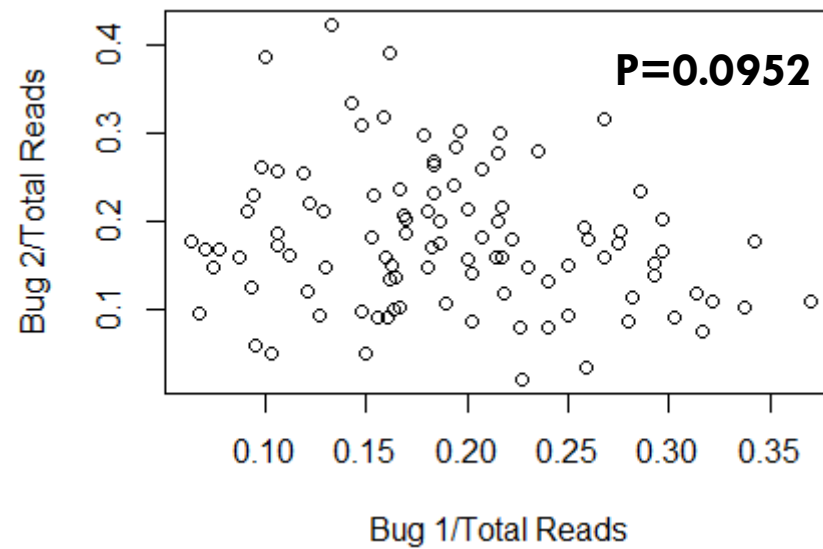
N = 3 taxa



N = 5 taxa



N = 10 taxa



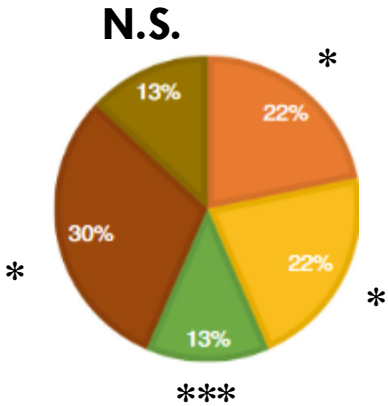
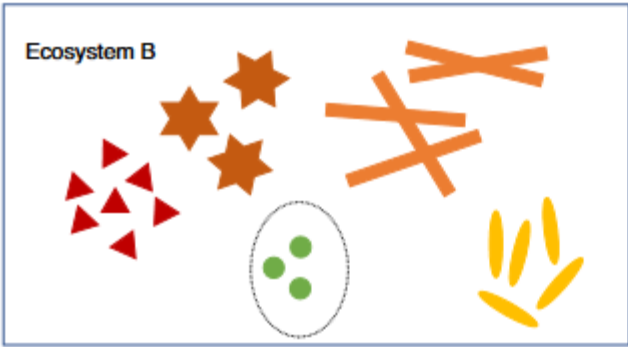
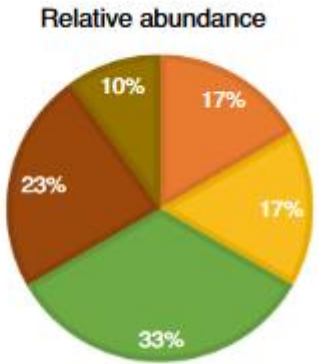
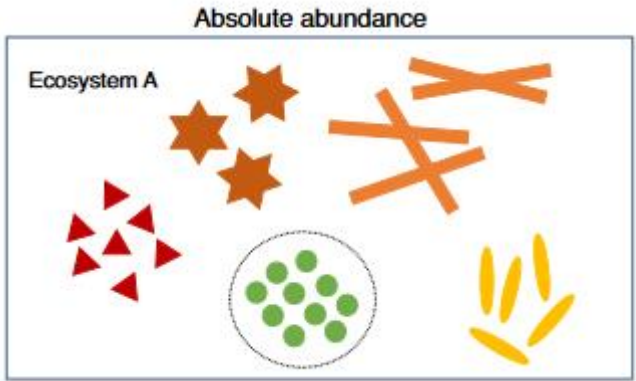
PITFALLS OF RELATIVE ABUNDANCE

Problem 2: Sequencing depths are strongly influenced by a small number of taxa (the **most abundant microbes**).

A **small number of microbes** accounts for the **majority of the reads**.

Variance in highly-abundant taxa cause **disproportionately high variance** in rarer taxa

High false positive rate



Colour	P value
Red	0.02
Brown	0.09
Orange	0.03
Yellow	0.04
Green	1.2e-13

Lin &
Peddada,
2020

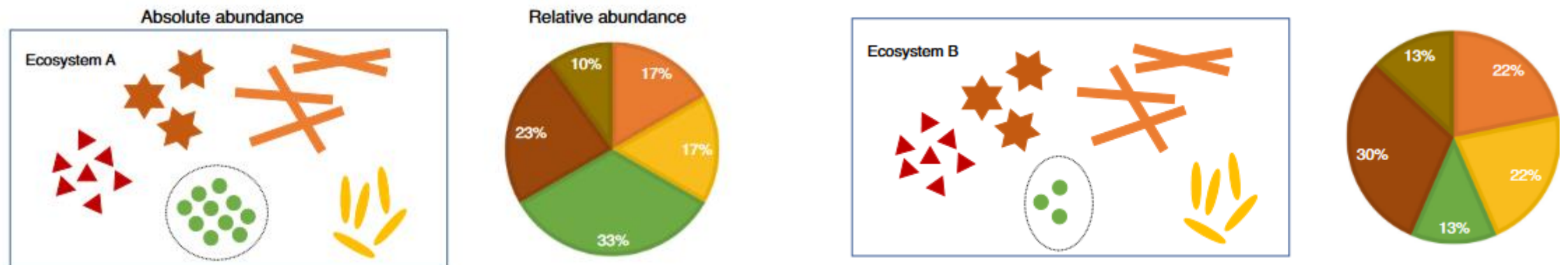
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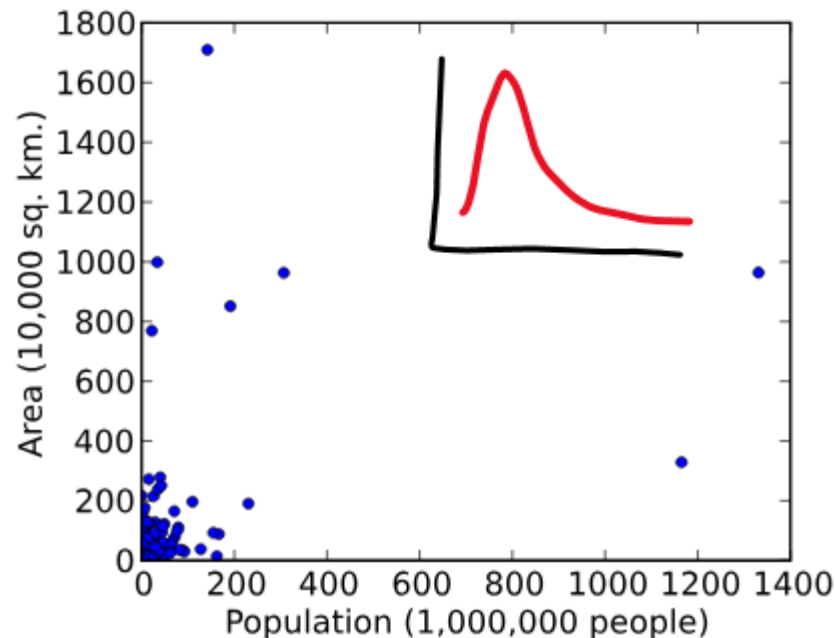
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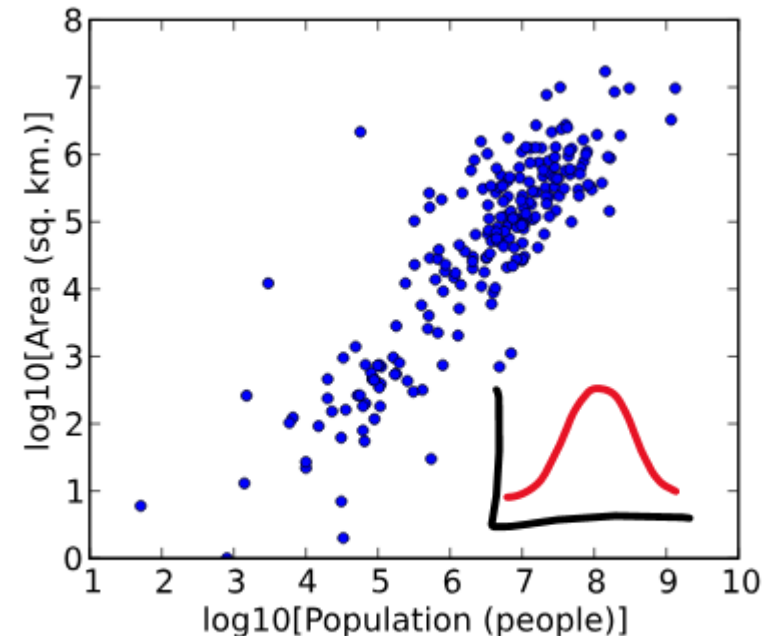


SIMPLE LOG TRANSFORMATIONS (LN, LOG₂, LOG₁₀, ETC)

- Fix right skewness
 - Useful approach for **negative binomial data**
- Reduces influence of **outliers**
- Simple log transformations **don't fix compositionality issues**



Log 10
→

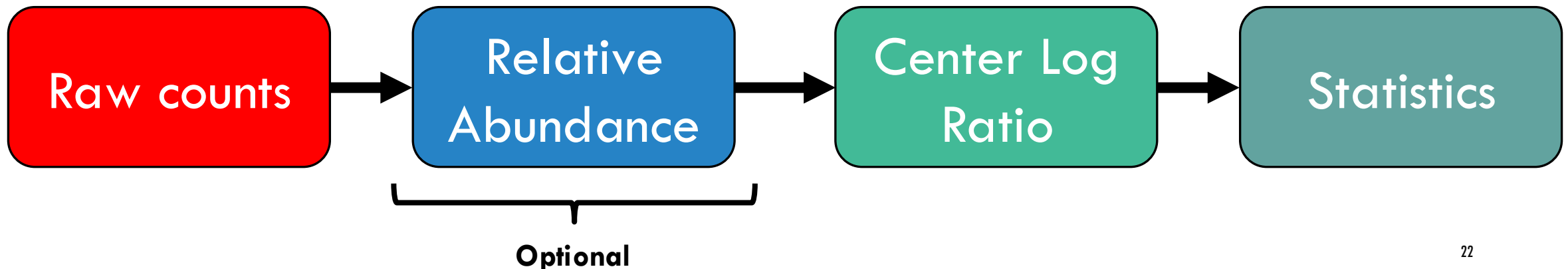


COMPOSITIONAL DATA ANALYSIS (CODA) METHODS: CENTERED LOG-RATIO (CLR)

Relative abundance: compositional dataset, every value affects every other value

- Traditional statistics are **not compatible**

CoDA methods are designed to **allow traditional statistics** to be applied to compositional (or absolute!) datasets



CLR IS THE MOST COMMONLY USED LOG RATIO IN MICROBIOME DATA.

For each sample, normalizes to **geometric mean of total reads** instead of **total reads**

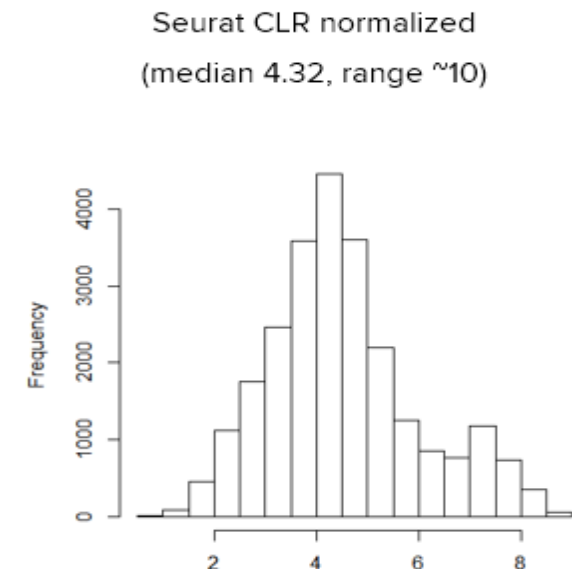
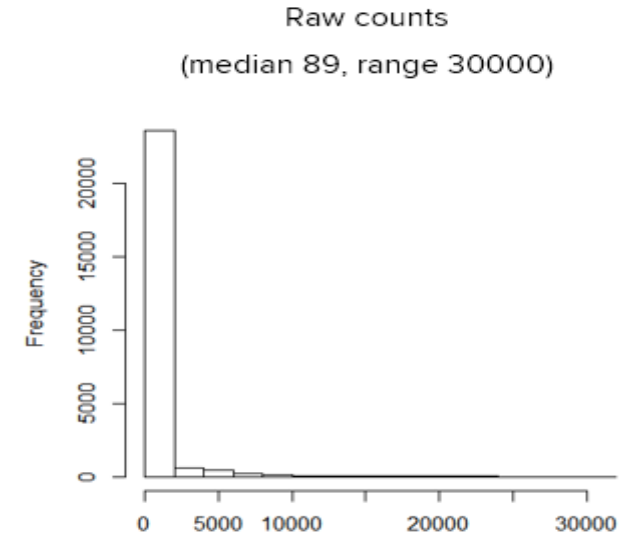
- Ex. GeoMean of [10, 100, 1000] = 100

Output is **less skewed** towards highly abundant taxa

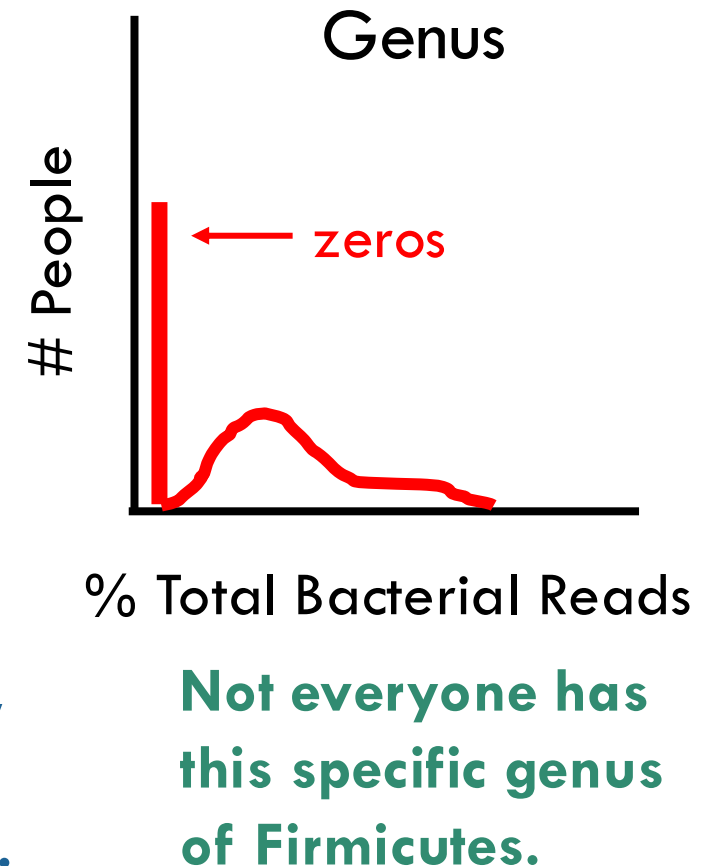
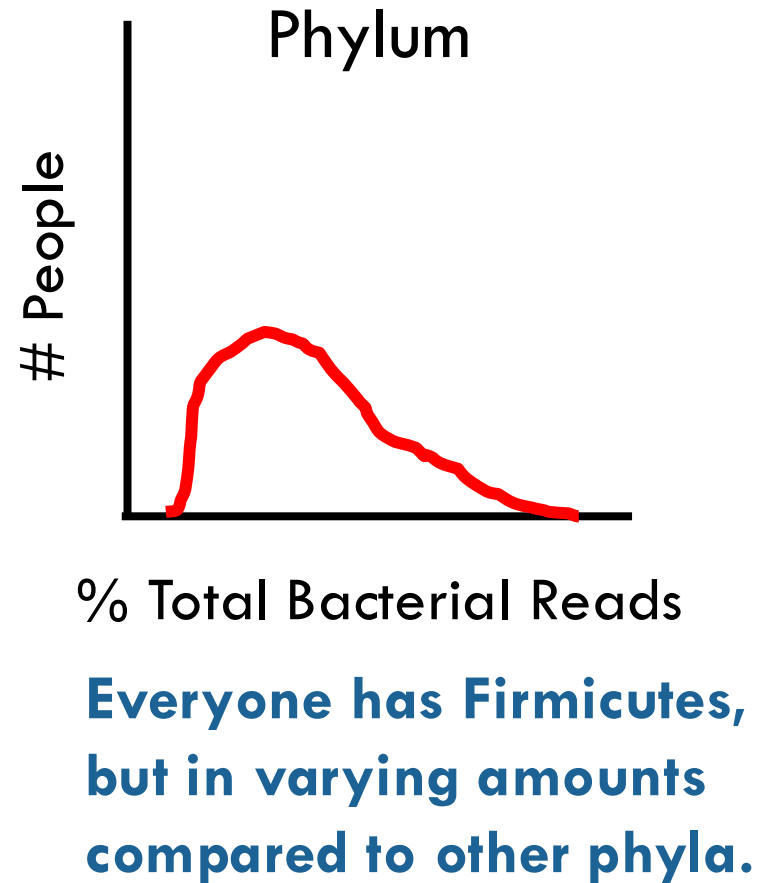
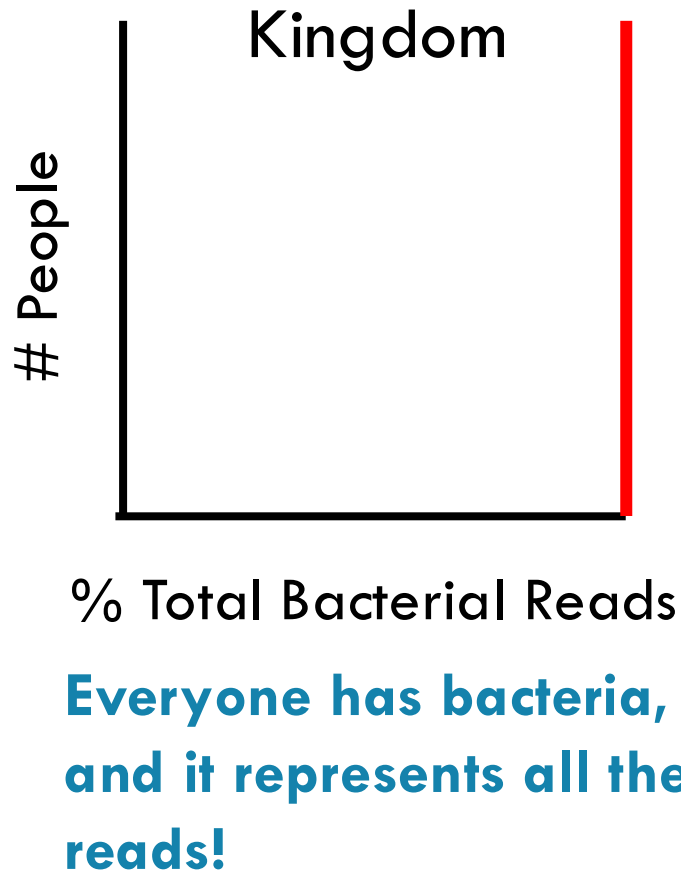
Already one of the default options for many tools

For each sample:

1. Find GeoMean of all values
2. Divide each value by the GeoMean
3. Take the natural log of each value



OVERABUNDANCE OF ZEROS: “SPARSENESS”



WHAT CAUSES ZERO ABUNDANCE?



1. True zero

Microbe is **not present** in sample.

2. Low abundance

Microbial abundance is below the **sequencing threshold**.

- Sequencing depth (# reads)
- Size of sampling fraction

3. Random chance

The sampling fraction is **disproportionate** compared to the 'true' microbiome.



Sampling fraction

THE PROBLEM WITH ZEROS

Can't be normalized to a normal distribution

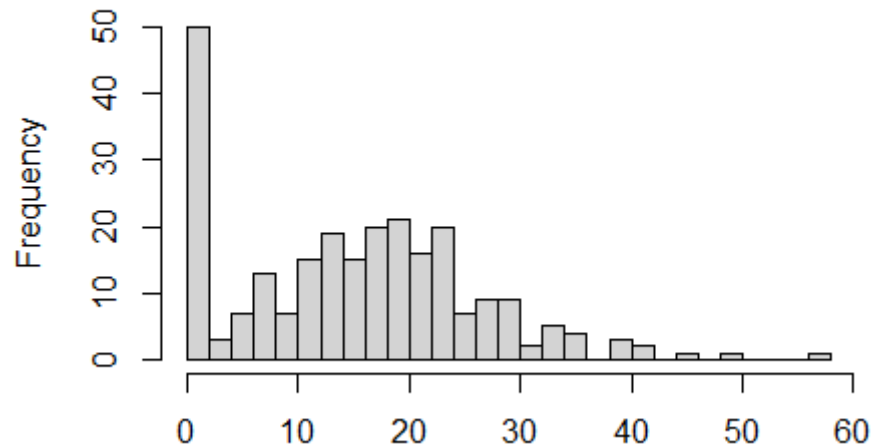
- Required for many statistical tests

Cannot be log normalized

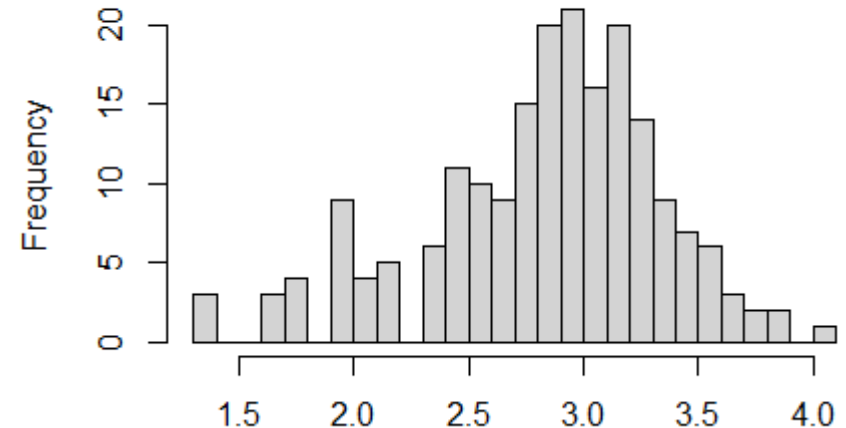
- $\text{Log}(0) = -\text{Infinity}$
- Automatically removed from plots in R, sometimes without warning

Sample	Abundance	Log
201	0	-Inf
202	0	-Inf
203	0	-Inf
204	0	-Inf
205	0	-Inf
206	0	-Inf
207	0	-Inf
208	0	-Inf

Histogram of pct_microbe\$Abundance



Histogram of pct_microbe\$Log



HOW TO DEAL WITH ZEROS:

Pseudocounts

Divide into present/absent

Choose a statistical method
that addresses zeros internally



Many differential abundance tools do this

PSEUDOCOUNTS: JUST ADD A REALLY TINY VALUE TO EVERYTHING

Raw counts: add count of 1

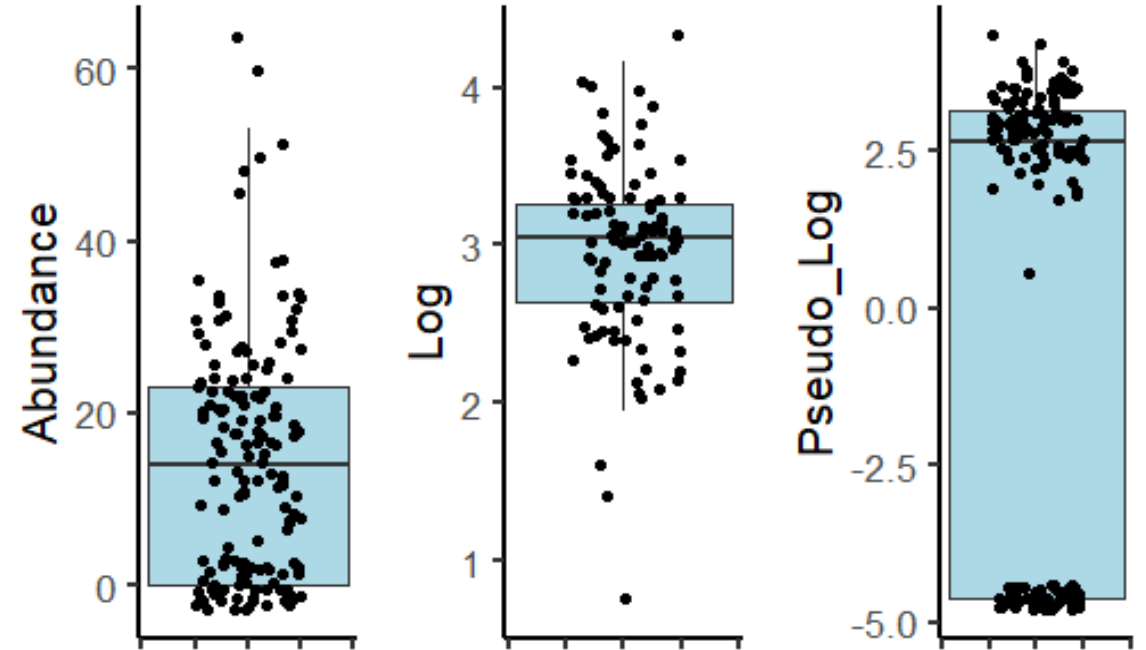
% abundance/other: variable, often $\frac{1}{2}$ the smallest nonzero value

Quick, easy, **very common**

Reasonable representation of data

Arbitrary value, still not possible to take normal distribution

- **Can skew statistics** in a manner dependent on the specific value of the pseudocount



TOPICS COVERED TODAY

1. Metagenomic sequencing principles
 1. How sequencing works
 2. Metagenomics pipeline
 3. Sequencing data distribution vs. the actual microbiome
 4. Data normalization
 5. Compositional Data Analysis: Log Transformations
 6. Overabundance of Zeros
2. Raw Data Processing (Tutorial)
3. Key R Analyses (Tutorial)

TOPICS COVERED TODAY

1. Metagenomic sequencing principles
- 2. Raw Data Processing (Tutorial)**
 1. Intro to Linux
 2. Check data quality (FASTQC)
 3. Clean and trim data (BBDUK)
 4. Remove human reads (HOCORT)
 5. Check data quality (FASTQC)
 6. Annotate taxonomy and function (BioBakery)
3. Key R Analyses (Tutorial)

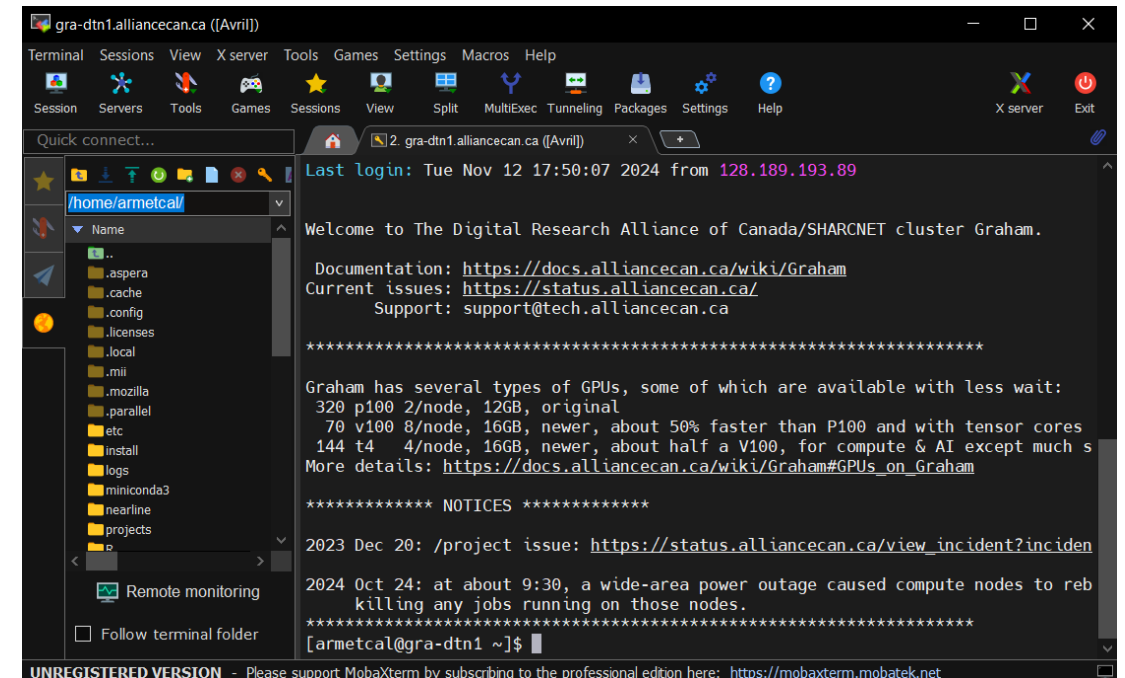
PROCESSING RAW DATA

- Raw reads are processed in **Linux** (command line/bash commands).
- Command line is **very fast** → good for large files
- Command line is **not optimized for statistics and visualization**
 - We will therefore **move to R** once we have organized our data into (relatively) small tables
- Huge data = **huge computational requirements**
 - Need to use supercomputers (Compute Canada, **ARC Sockeye**, etc)



ACCESSING SUPERCOMPUTER RESOURCES

- Can access via your computer's command line or by **using downloadable programs**
 - Programs are more user friendly
- Your computer tells the supercomputer what to do
 - The computational load is given to the supercomputer
- Recommended Windows program: **MobaXTerm**
- Recommended Mac program: **Terminus, Tabby Terminal, iTerm2**

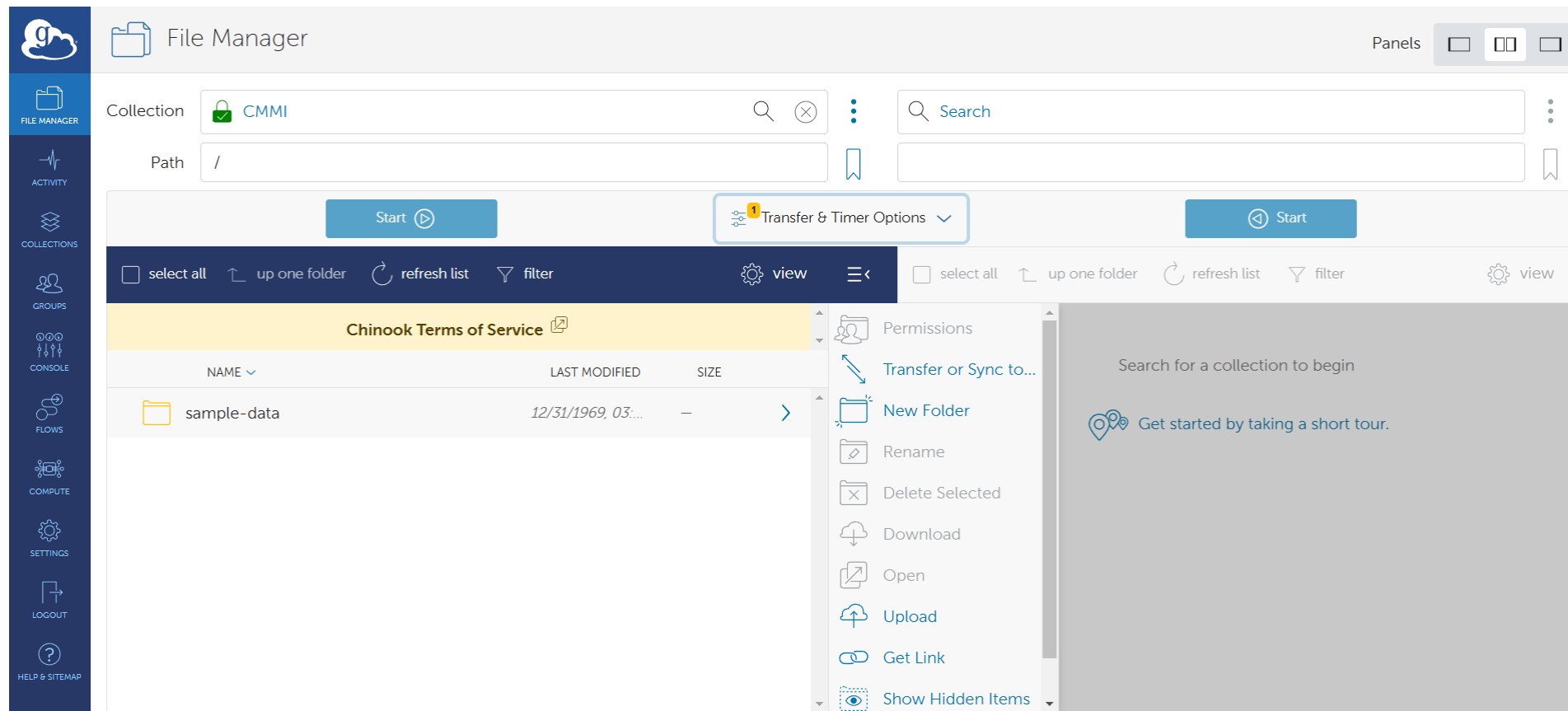


```
gra-dtn1.alliancecan.ca (Avril)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/home/armetcal/
Name
..
.aspera
.cache
.config
.licenses
.local
.mil
.mozilla
.parallel
.etc
.install
.logs
.miniconda3
.nearline
.projects
.e
Remote monitoring
Follow terminal folder
Last login: Tue Nov 12 17:50:07 2024 from 128.189.193.89
Welcome to The Digital Research Alliance of Canada/SHARCNET cluster Graham.
Documentation: https://docs.alliancecan.ca/wiki/Graham
Current issues: https://status.alliancecan.ca/
Support: support@tech.alliancecan.ca
*****
Graham has several types of GPUs, some of which are available with less wait:
320 pi100 2/node, 12GB, original
70 v100 8/node, 16GB, newer, about 50% faster than P100 and with tensor cores
144 t4 4/node, 16GB, newer, about half a V100, for compute & AI except much s
More details: https://docs.alliancecan.ca/wiki/Graham#GPUs\_on\_Graham
***** NOTICES *****
2023 Dec 20: /project issue: https://status.alliancecan.ca/view\_incident?inciden
2024 Oct 24: at about 9:30, a wide-area power outage caused compute nodes to reb
killing any jobs running on those nodes.
*****
[armetcal@gra-dtn1 ~]$
```

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>

TRANSFER FILES BETWEEN SYSTEMS USING GLOBUS

<https://app.globus.org/>



LINUX MUST-KNOWS

- Navigating file systems:
 - **ls** → list files in current folder
 - **cd** → change directory
 - **..** → go up one directory (ex. **cd ..** , **cd ../file/path**)
- The majority of metagenomics-related packages use **anaconda** (conda) to aid installation
 - This will make installing programs MUCH easier
 - Compute Canada comes with many packages pre-installed and prefers if you don't use conda
 - Install the mini version, **Miniconda**, here: <https://docs.anaconda.com/miniconda/>



https://docs.alliancecan.ca/wiki/Available_software

Main Resources @ AC CC Notion Guitar Resources Misc. BV Streaming Video Re... UVC Audio Cine

List of globally-installed modules [\[edit\]](#)

The table below lists software for which an environment module has been installed on our systems. In simple cases, the command to configure your environment. In more complicated cases, some prerequisite modules may also need to be prerequisites and brief notes about the software. If more extensive documentation about a package is available, there v to sort in ascending or descending order for a given column. In particular, sorting by software type might be of interest. bioinformatics), chem (chemistry), geo (earth sciences), io (input/output), math (mathematics), mpi (**MPI**), phys (physics

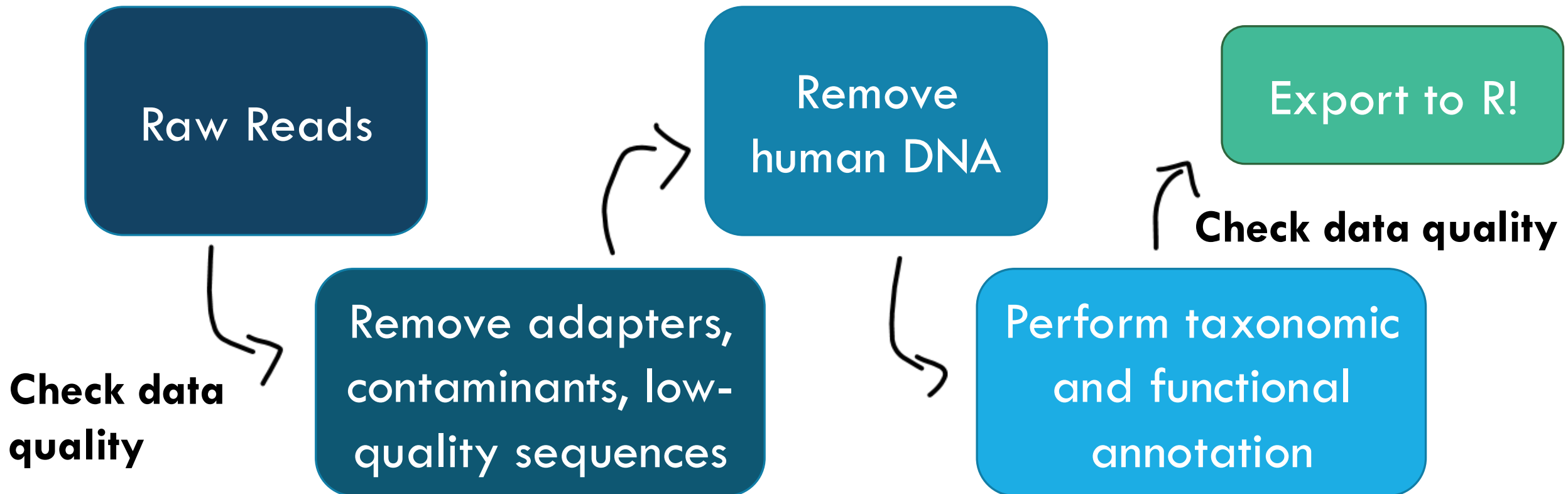
AVX512 (Béluga, Niagara, Cedar/Graham Skylake) AVX2 (Cedar/Graham Broadwell) AVX SSE3

Module	Type	Versions
4ti2	-	1.6.9
abaqus	phys	2021
abinit	chem	9.2.2, 9.6.2, 10.0.3
abricate	-	1.0.0

Command line: **module spider search-term**

TUTORIAL: RAW DATA PIPELINE

<https://armetcal.github.io/Bioinformatic-Tool-Wikipedia/> (Name to be updated in future)



FASTQC/MULTIQC: ASSESS DATA QUALITY

- <https://anaconda.org/bioconda/fastqc>
- Tests sequencing data for a number of potential issues
 - Can check and compare quality at each processing step
- Each sample generates a separate HTML file
- Arduous to check every file
- **MultiQC:** results presented in tabular format
 - All samples together
 - Easier to parse, but no plots
 - <https://anaconda.org/bioconda/multiqc>

FastQC Report

Summary

- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ✗ [Per base sequence content](#)
- ! [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ✓ [Sequence Length Distribution](#)
- ✗ [Sequence Duplication Levels](#)
- ✓ [Overrepresented sequences](#)
- ✓ [Adapter Content](#)

REMOVING UNWANTED SEQUENCES



Relevant portion: 0:40-1:29

REMOVING UNWANTED SEQUENCES (SOME KEY EXAMPLES)

Normal sequence: remove adapters only

ATCGTAGCTAGCTAGCATCGATGCATCGTAGCTATCTATGCTGACT

Monomeric repeats: complexity is too low

ATCGTAGCTAGCTTTTTTTTTTTAAAAAAATCTATGCTGACT

Low-quality reads: bases cannot be called

ATCGTAGCTAGCBVIBIVSDFVBUKYADNCDHNNCTCTATGCTGACT

Sequencing controls/other known contaminants: no useful information

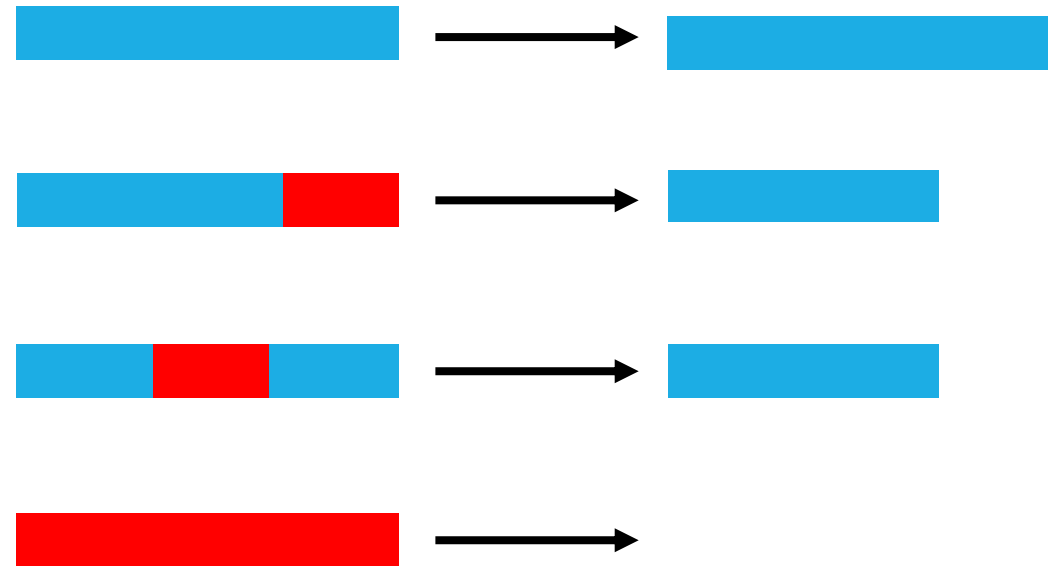
ATCGTAGCTAGCATCGATCGATCGATCGATCTCTATGCTGACT

BBTOOLS PACKAGE: BBDUK

FILTER (ALMOST) EVERYTHING IN ONE STEP!

<https://jgi.doe.gov/data-and-tools/software-tools/bbtools/bbtools-user-guide/bbduk-guide/>

- Combines most common data-quality-related trimming, filtering, and masking operations (huge number of options)
- Most options can be run concurrently, with very few exceptions



BASIC IMPLEMENTATION OF BBDOUK

```
bbduk.sh \
```

* If single-end sequencing,
remove the in2 and out2 lines

```
in="forward_read.fastq.gz" \
```

```
in2="reverse_read.fastq.gz" \
```

} Same inputs as FastQC

```
out=bbduk_out/"forward_read.fastq.gz" \
```

```
out2=bbduk_out/"reverse_read.fastq.gz" \
```

} Save location/name
of filtered samples

```
stats=bbduk_out/stats/"read_stats.txt" \
```

} Output that summarizes
trimmed samples

```
tpe tbo qtrim=rl trimq=25 maq=25 minlen=50 ref=adapters,phix \
```

```
-Xmx47g
```

} Set to slightly lower than your total
available memory (ex. 50G → 47G)

All trimming parameters



RUNNING ON MULTIPLE SAMPLES

```
for file in sequences/*_1.fastq.gz; do
    sample=$(basename "$file" "_1.fastq.gz") # Extracts the base sample name
    bbduk.sh \
        in="${sample}_1.fastq.gz" \
        in2="${sample}_1.fastq.gz" \
        out=bbduk_out/"${sample}_1.fastq.gz" \
        out2=bbduk_out/"${sample}_1.fastq.gz" \
        stats=bbduk_out/stats/"${sample}_stats.txt" \
        tpe tbo qtrim=rl trimq=25 maq=25 minlen=50 ref=adapters,phix \
        -Xmx47g
done
```

BASIC TRIMMING PARAMETERS

```
bbduk.sh \  
  in="${sample}_1.fastq.gz" \  
  in2="${sample}_1.fastq.gz" \  
  out=bbduk_out/"${sample}_1.fastq.gz" \  
  out2=bbduk_out/"${sample}_1.fastq.gz" \  
  stats=bbduk_out/stats/"${sample}_stats.txt" \  
  tpe tbo qtrim=r1 trimq=25 maq=25 minlen=50 ref=adapters,phix \  
  -Xmx47g
```

BASIC TRIMMING PARAMETERS

```
tpe tbo qtrim=r1 trimq=25 maq=25 minlen=50 ref=adapters,phix
```

tpe → Forward and reverse reads trimmed to same length

tbo → Optimize adapter trimming

qtrim=r1 → Quality trim both the right and left sides of the read

trimq=25 → Quality trim edges to a min Phred quality score of 25

maq=25 → Remove reads with mean quality score <25

minlen=50 → Remove reads that have <50 bp after trimming

ref=adapters,phix → In-built libraries of common adapter sequences and contaminants. Matching sequences will be removed from reads

- In this case, reads contain generic adapters and PhiX quality control reads, which will be removed.

ADAPTERS & CONTAMINANTS (REF=ADAPTERS,PHIX)

- BBDuk has large variety of contaminant lists that can be used
 - <https://github.com/BioInfoTools/BBMap/tree/master/resources>
- 'Adapters' includes most common next-gen sequencing adapters
 - <https://github.com/BioInfoTools/BBMap/blob/master/resources/adapters.fa>
- PhiX: common QC DNA used during sequencing
- If using pre-made list, matches to ANY of the sequences in the list will be removed
 - Usually sequences aren't microbial, so not a problem
 - Still good to be aware
 - Can create your own list if desired

```
>I5_Nextera_Transposase_1
CTGTCTCTTATACACATCTGACGCTGCCGACGA
>I7_Nextera_Transposase_1
CTGTCTCTTATACACATCTCCGAGCCCACGAGAC
>I5_Nextera_Transposase_2
CTGTCTCTTATACACATCTCTGATGGCGCGAGGGAGGC
>I7_Nextera_Transposase_2
CTGTCTCTTATACACATCTCTGAGCGGGCTGGCAAGGC
>I5_Primer_Nextera_XT_and_Nextera_Enrichment_[N/S/E]501
GACGCTGCCGACGAGCGATCTAGTGTAGATCTCGGTGGTCGCCGTATCATT
```

Part of adapters.fa

OUTPUTS OF BBDOUK

1. **Log** file/console output

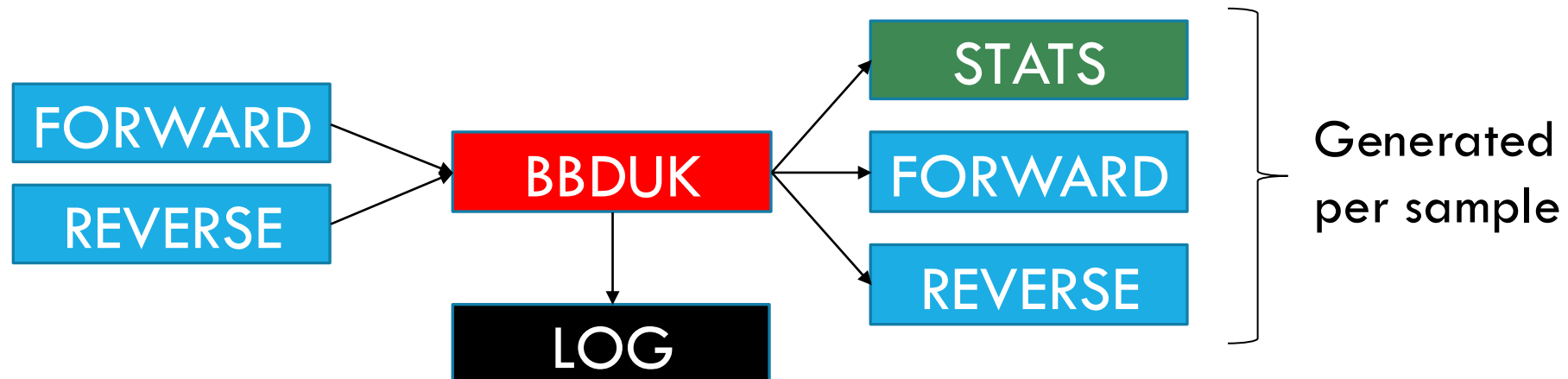
1. One generated per run
2. Contains summary information on each sample included in run, printed sequentially

2. **FASTQ** sequencing files

1. Two separate files per sample (1 forward, 1 reverse)

3. **Stats** files

1. One file per sample; summarizes adapter/contaminant removal



CRITICAL: CHECK FOR SUCCESS

- BBDuk has a high CPU/memory requirement
- Common to encounter errors
 - Not all samples have trim summaries (red)
 - Visible error (ex. bottom image)
- However, **NOT ALWAYS CLEAR** whether an error has occurred at first glance
- This is because **output fastq files are still generated for failed samples**
 - However, stats files are NOT generated

Success

```
java -ea -Xmx45g -Xms45g -cp /cvmfs/soft.computecanada.ca/easybuild/software/2023/x86-64-v4
Executing jgi.BBDuk [in=sequences/SRR19064990_1.fastq.gz, in2=sequences/SRR19064990_2.fastq.gz]
Version 39.06

0.131 seconds.
Initial:
Memory: max=48318m, total=48318m, free=48216m, used=102m

Added 8330 kmers; time: 0.095 seconds.
Memory: max=48318m, total=48318m, free=48065m, used=253m

Input is being processed as paired
Started output streams: 0.380 seconds.
Processing time: 129.378 seconds.

Input: 45404264 reads 6856043864 bases.
Contaminants: 1775984 reads (3.91%) 268173584 bases (3.91%)
QTrimmed: 22949892 reads (50.55%) 1978285095 bases (28.85%)
Trimmed by overlap: 1468058 reads (3.23%) 18568496 bases (0.27%)
Low quality discards: 0 reads (0.00%) 0 bases (0.00%)
Low entropy discards: 9240 reads (0.02%) 1161622 bases (0.02%)
Total Removed: 11892142 reads (26.19%) 2266188797 bases (33.05%)
Result: 33512122 reads (73.81%) 4589855067 bases (66.95%)

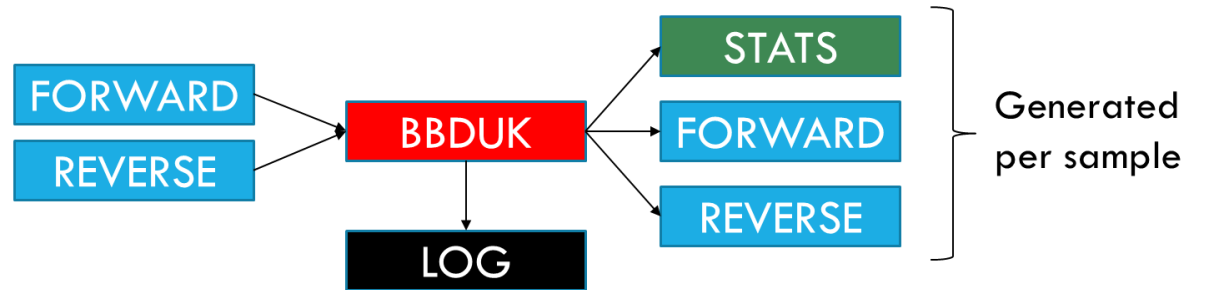
Time: 129.935 seconds.
Reads Processed: 45404k 349.44k reads/sec
Bases Processed: 6856m 52.77m bases/sec
```

Failure

```
Input: 45404264 reads 6856043864 bases.
Contaminants: 1775984 reads (3.91%) 268173584 bases (3.91%)
QTrimmed: 22949892 reads (50.55%) 1978285095 bases (28.85%)
Trimmed by overlap: 1468058 reads (3.23%) 18568496 bases (0.27%)
Low quality discards: 0 reads (0.00%) 0 bases (0.00%)
Low entropy discards: 9240 reads (0.02%) 1161622 bases (0.02%)
Total Removed: 11882902 reads (26.17%) 2265027175 bases (33.04%)
Result: 33521362 reads (73.83%) 4591016689 bases (66.96%)

Time: 326.662 seconds.
Reads Processed: 45404k 138.99k reads/sec
Bases Processed: 6856m 20.99m bases/sec
slurmstepd: error: Detected 1 oom_kill event in StepId=51559701.batch. Some of the step tasks have been OOM Killed.
```

ORDER OF OPERATIONS



1. Check log file: most should have trim summaries, no error messages
2. Check that:
 1. # Fastq files in = # Fastq files out
 2. # Fastq files = 2*# Stats files
3. If the number of stats files is LESS THAN half the number of Fastq files, some samples failed.
 1. Ex. # Fastq = 60, # stats = 28 → 2 samples failed
 2. [Follow these troubleshooting instructions](#)

BBDUK OUTPUTS

- **Log file:** summarizes all filtered data
 - Common to have 60-80% reads remaining, if data quality is reasonable
- **Fastq files:** will be smaller than inputs
 - Roughly proportional to the amount of data filtered out
- **Stats file:** Known adapters/contaminants should constitute majority of the removed sequences

Log File: Trimming and Quality Filtering

```
Input: 45404264 reads 6856043864 bases.
Contaminants: 1775984 reads (3.91%) 268173584 bases (3.91%)
QTrimmed: 22949892 reads (50.55%) 1978285095 bases (28.85%)
Trimmed by overlap: 1468058 reads (3.23%) 18568496 bases (0.27%)
Low quality discards: 0 reads (0.00%) 0 bases (0.00%)
Low entropy discards: 9240 reads (0.02%) 1161622 bases (0.02%)
Total Removed: 11892142 reads (26.19%) 2266188797 bases (33.05%)
Result: 33512122 reads (73.81%) 4589855067 bases (66.95%)

Time: 129.935 seconds.
Reads Processed: 45404k 349.44k reads/sec
Bases Processed: 6856m 52.77m bases/sec
```

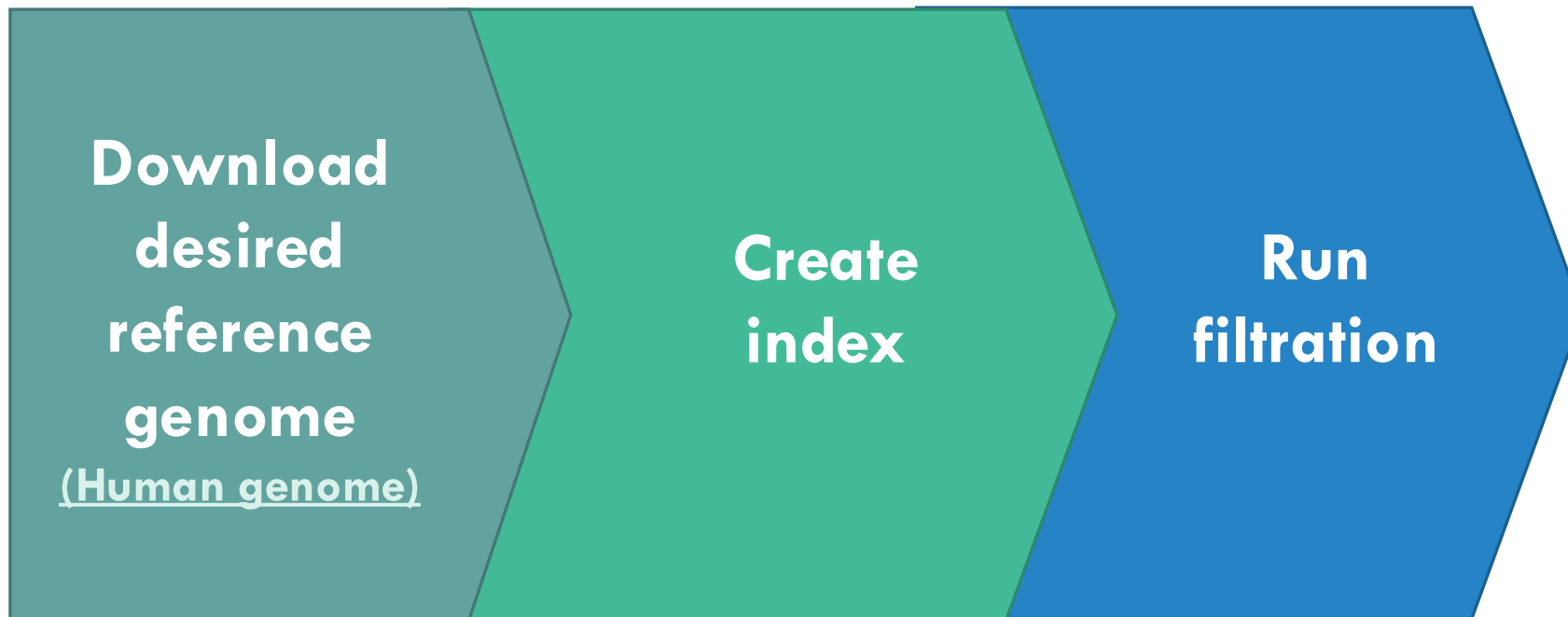
Stats File: Removed Sequences

```
GNU nano 7.2 SRR19064990_stats.txt
#File sequences/SRR19064990_1.fastq.gz sequences/SRR19064990_2.fastq.gz
#Total 45404264
#Matched 1735456 3.82223%
#Name Reads ReadsPct
I5_Nextera_Transposase_1 866825 1.90913%
I7_Nextera_Transposase_1 853593 1.87998%
I5_Primer_Nextera_XT_Index_Kit_v2_S511 8235 0.01814%
I7_Primer_Nextera_XT_and_Nextera_Enrichment_N707 6588 0.01451%
TruSeq_Universal_Adapter 63 0.00014%
PhiX_read2_adapter 35 0.00008%
TruSeq_Adapter_Index_11 24 0.00005%
I5_Primer_Nextera_XT_and_Nextera_Enrichment_[N/S/E]503 14 0.00003%
I5_Primer_Nextera_XT_and_Nextera_Enrichment_[N/S/E]502 13 0.00003%
I5_Primer_Nextera_XT_Index_Kit_v2_S510 12 0.00003%
I7_Primer_Nextera_XT_Index_Kit_v2_N718 7 0.00002%
I5_Primer_Nextera_XT_and_Nextera_Enrichment_[N/S/E]506 6 0.00001%
I5_Primer_Nextera_XT_and_Nextera_Enrichment_[N/S/E]507 6 0.00001%
I5_Primer_Nextera_XT_and_Nextera_Enrichment_[N/S/E]504 4 0.00001%
I7_Primer_Nextera_XT_and_Nextera_Enrichment_N701 3 0.00001%
PhiX_read1_adapter 3 0.00001%
I5_Primer_Nextera_XT_Index_Kit_v2_S516 2 0.00000%
```

HOCORT: REMOVE HOST READS

- Any metagenomic dataset will have reads that map to the host
- Can technically be performed using BBTools as well, but is less robust
- HoCoRT: Host Contamination Remover Tool
- Compares each read to a reference genome, removes any matches
 - Can also set it to keep matches & discard anything else

HOCORT PIPELINE



BUILDING A HOCORT INDEX

- Reference genome is converted into a series of 'index' datasets
 - Essentially a 'dictionary' written in HoCoRT's native language
- Input is reference genome, NOT samples
- Output will create a list of index files with the prefix 'human.index'

NAME ▾	LAST MODIFIED	SIZE
 human.index.1.bt2	10/2/2024, 12...	1.04 GB
 human.index.2.bt2	10/2/2024, 12...	784.20 MB
 human.index.3.bt2	10/2/2024, 12...	17.56 KB
 human.index.4.bt2	10/2/2024, 12...	784.20 MB
 human.index.rev.1.bt2	10/2/2024, 12...	1.04 GB
 human.index.rev.2.bt2	10/2/2024, 12...	784.20 MB

```
hocort index bowtie2 \  
--input GCF_000001405.40_GRCh38.p14_genomic.fna.gz \  
--output full/filepath/to/index/human.index
```

RUNNING HOCORT

hocort map bowtie2 \

-x full/path/to/index/human.index \ **# Same as output from building index**

-i bbduk_out/file_1.fastq.gz bbduk_out/file_2.fastq.gz \ **# Two FASTQ files**

-o hocort_out/file_1.fastq.gz hocort_out/file_2.fastq.gz \ **# Two FASTQ files**

--filter true **# REMOVE any matches (true=keep matches)**

```
GNU nano 7.2 slurm-51589744.out
/localscratch/spool/slurmd/job51589744/slurm_script: line 12: activate: No such file or directory
2024-10-02 15:54:45,825 INFO | Running pipeline: Bowtie2
2024-10-02 16:09:09,685 INFO | 12596003 reads; of these:
2024-10-02 16:09:10,210 INFO | 12596003 (100.00%) were paired; of these:
2024-10-02 16:09:10,210 INFO | 12595849 (100.00%) aligned concordantly 0 times
2024-10-02 16:09:10,210 INFO | 21 (0.00%) aligned concordantly exactly 1 time
2024-10-02 16:09:10,210 INFO | 133 (0.00%) aligned concordantly >1 times
2024-10-02 16:09:10,210 INFO | ----
2024-10-02 16:09:10,210 INFO | 12595849 pairs aligned concordantly 0 times; of these:
2024-10-02 16:09:10,210 INFO | 3 (0.00%) aligned discordantly 1 time
2024-10-02 16:09:10,210 INFO | ----
2024-10-02 16:09:10,210 INFO | 12595846 pairs aligned 0 times concordantly or discordantly; of these:
2024-10-02 16:09:10,210 INFO | 25191692 mates make up the pairs; of these:
2024-10-02 16:09:10,210 INFO | 25190371 (99.99%) aligned 0 times
2024-10-02 16:09:10,210 INFO | 22 (0.00%) aligned exactly 1 time
2024-10-02 16:09:10,210 INFO | 1299 (0.01%) aligned >1 times
2024-10-02 16:09:10,210 INFO | 0.01% overall alignment rate
2024-10-02 16:09:10,276 INFO | [M::bam2fq_mainloop] discarded 0 singletons
2024-10-02 16:09:10,276 INFO | [M::bam2fq_mainloop] processed 25189104 reads
2024-10-02 16:09:10,726 INFO | Pipeline Bowtie2 run time: 864.7799396514893 seconds
2024-10-02 16:09:10,726 INFO | Pipeline exited with returncode: 0
```

LAST STEP: RERUN FASTQC

[FastQC website](#)

[Help Pages](#)

- FASTQC: Checks for variety of sequencing data issues
- Of errors noted in raw files, majority should be resolved/reduced to warnings
 - Note: it is common for “Per Base Sequence Content” to fail and is not necessarily problematic.
- Inspect any remaining issues, refine approach if needed

Overrepresented sequences

Sequence	Count	Percentage	Possible Source
GAGATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCTCGTATGCC	25205	0.3219146546202512	Illumina Paired End PCR Primer 2 (100% over 49bp)



 **Overrepresented sequences**
No overrepresented sequences

**SEQUENCING DATA ARE NOW CLEAN:
READY FOR ANNOTATION!**

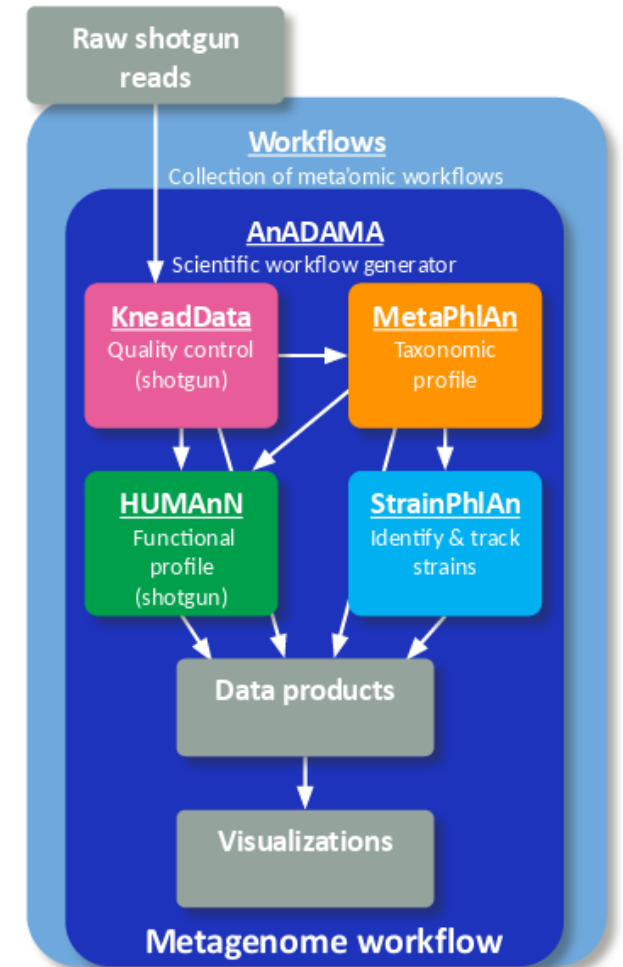
TOPICS COVERED TODAY

1. Metagenomic sequencing principles
2. **Raw Data Processing (Tutorial)**
 1. Intro to Linux
 2. Check data quality (FASTQC)
 3. Clean and trim data (BBDUK)
 4. Remove human reads (HOCORT)
 5. Check data quality (FASTQC)
 6. **Annotate taxonomy and function (BioBakery)**
3. Key R Analyses (Tutorial)

BIOBAKERY: TECHNICALLY ALL YOU NEED

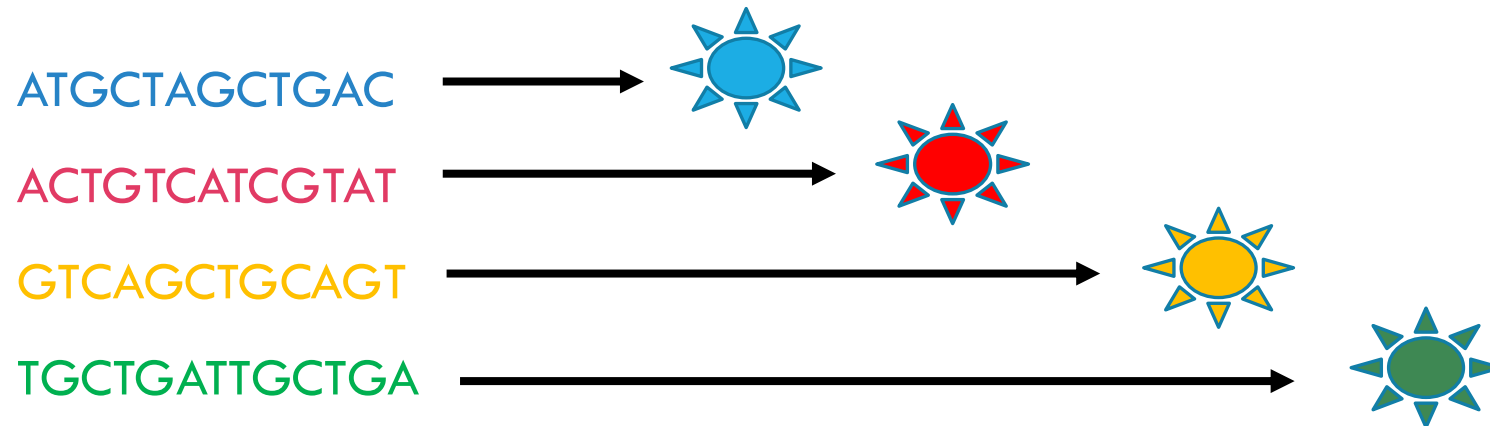
https://huttenhower.sph.harvard.edu/biobakery_workflows/

- KneadData: similar to QC pipeline presented here
 - Less commonly cited
- **MetaPhlAn: taxonomic annotation**
- **HUMAnN: functional annotation**
- StrainPhlAn: strain-level information

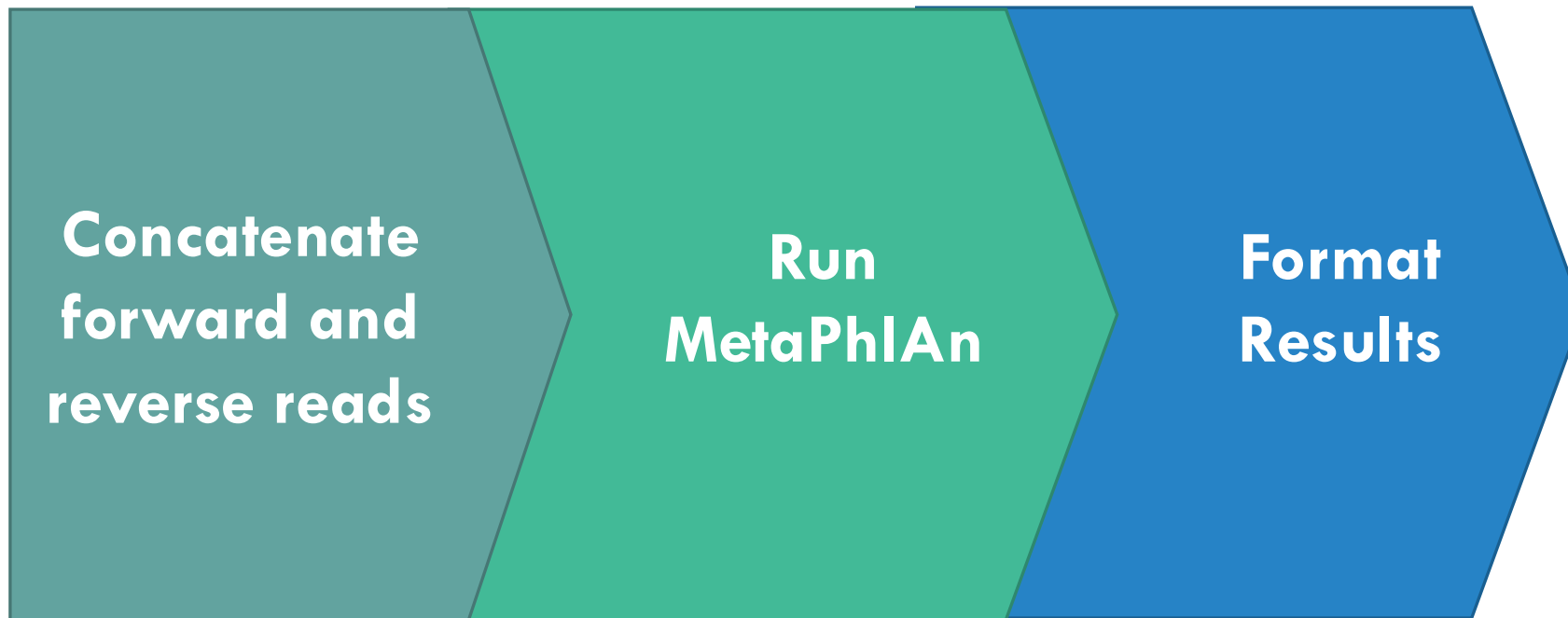


METAPHLAN: TAXONOMY

- Aligns reads to huge database of microbial DNA to assign taxonomy to **species level**
- Good quality results, relatively easy to use
- Prefers to format results using relative abundance, not counts
 - Recall: this is **problematic for many downstream analyses**
- Stay tuned for details

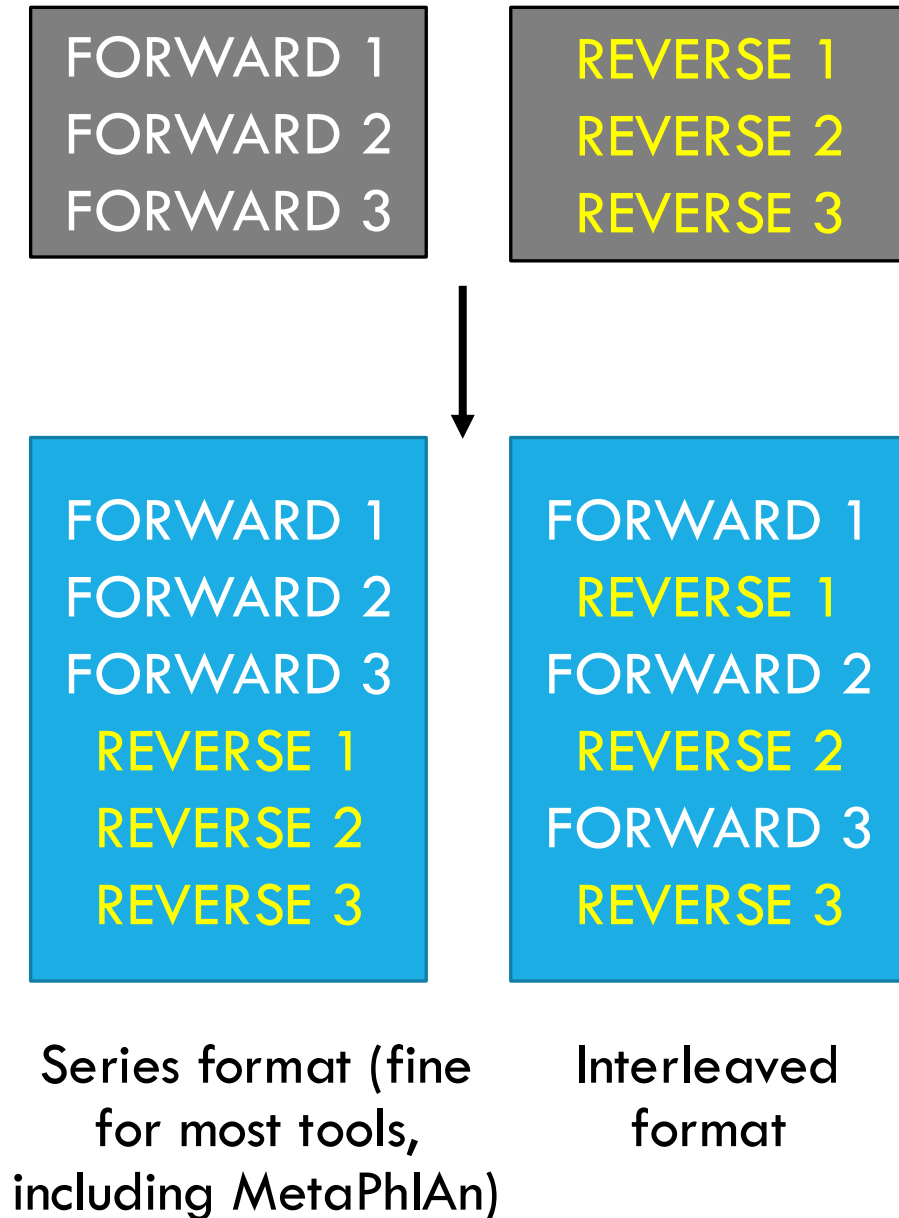


METAPHLAN PIPELINE



CONCATENATING READS

- By default, MetaPhlAn will only use forward reads if two Fastq files are provided
- Reads can be concatenated into a single Fastq file to avoid this
- [See this page for coding help](#)



RUNNING METAPHLAN

```
for file in sequences/*_1.fastq.gz; do
    sample=$(basename "$file" "_1.fastq.gz")
    metaphlan sequences/concatenated/"${sample}_1.fastq.gz" \
        --input_type fastq \
        --unclassified_estimation \ # How many reads don't align to any microbe?
        -o metaphlan_out/"${sample}.txt" \ # Taxonomy table output
        --nproc 14 \ # Slightly less than total available CPUs
        -t rel_ab_w_read_stats \ # Tabulate taxonomy in relative AND absolute counts
        --index mpa_vOct22_CHOCOPhlAnSGB_202403 \ # MetaPhlAn database
        --bowtie2db $SCRATCH/metaphlan_databases \ # Location of database
        # Below: save intermediate (makes reannotation very fast)
        --bowtie2out metaphlan_out/bt_out/"${sample}.bowtie2.txt"
done
```

INTERLUDE: COUNTS AND SIMILAR CONCEPTS

16S amplicon sequencing

All DNA fragments are from the same gene region

All amplicons are the same length

Each amplicon has an equal chance of being sequenced

Each taxonomy assignment can be represented as a count of 1

Whole metagenome sequencing

DNA fragments are from different genes

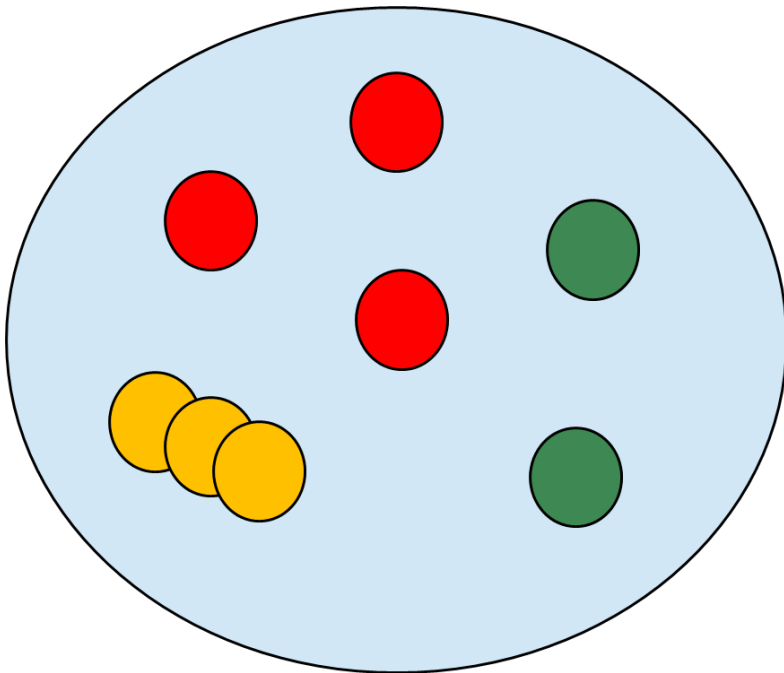
Gene lengths are highly variable

Longer genes are more likely to be detected by the sequencer

Raw counts are therefore biased toward long genes

Counts are therefore normalized to gene length (typically done automatically by software) → RPK, reads per kilobase

EXAMPLE: SEQUENCING 6 DNA MOLECULES



COUNTS

# Fragments:	3	3	2
% Fragments:	38	38	24

RPK

Norm. by length:	1	3	2
% Norm:	17	50	33

1 circle = 1 sequencing unit
circles = molecule length
Colour = unique DNA

Many variations of RPK – just know that they are analogous to counts and have their own statistical implications.

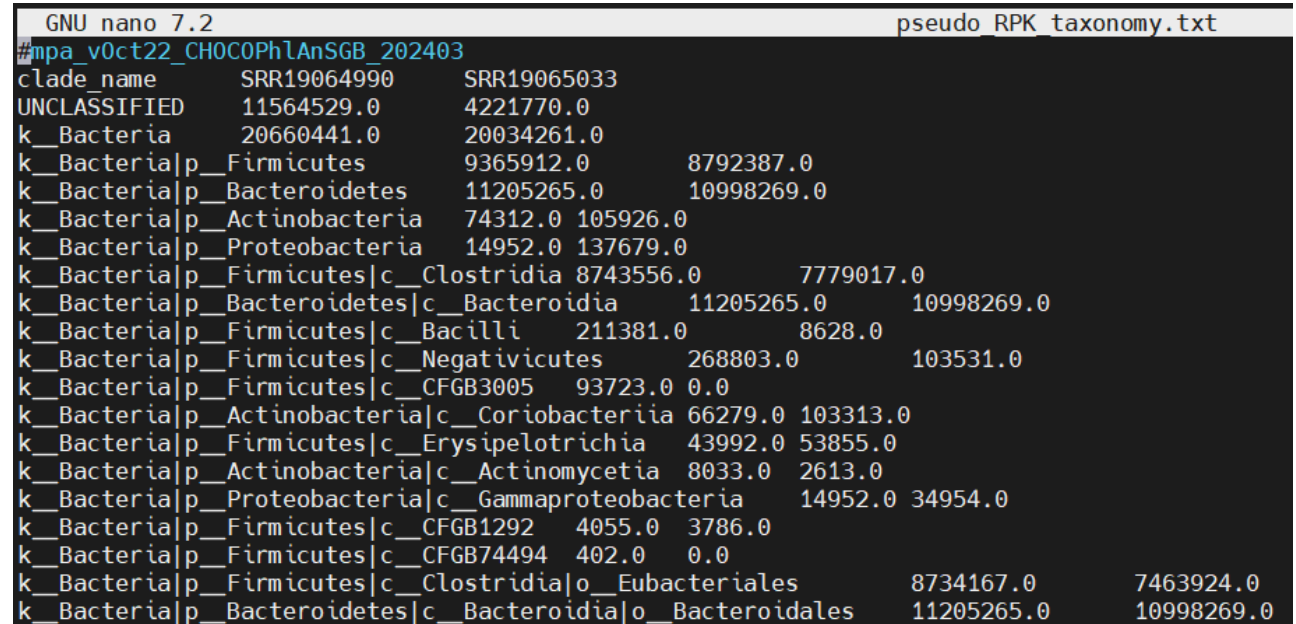
FORMAT THE RESULTS

This dataset can be input into R!

Basic output: 1 file per sample

Desired output: one big table

- Built-in command: combines all relative abundance data, but NOT count data
- Workaround: command can be easily modified to use the count (pseudo-RPK) data from 'rel_ab_with_stats' data
 - [Follow these instructions](#)



```
GNU nano 7.2 pseudo_RPK_taxonomy.txt
#mpa_v0ct22_CHOCOPhlAnSGB_202403
clade_name      SRR19064990      SRR19065033
UNCLASSIFIED    11564529.0        4221770.0
k_Bacteria      20660441.0        20034261.0
k_Bacteria|p_Firmicutes      9365912.0        8792387.0
k_Bacteria|p_Bacteroidetes  11205265.0        10998269.0
k_Bacteria|p_Actinobacteria  74312.0  105926.0
k_Bacteria|p_Proteobacteria  14952.0  137679.0
k_Bacteria|p_Firmicutes|c_Clostridia  8743556.0        7779017.0
k_Bacteria|p_Bacteroidetes|c_Bacteroidia  11205265.0        10998269.0
k_Bacteria|p_Firmicutes|c_Bacilli  211381.0        8628.0
k_Bacteria|p_Firmicutes|c_Negativicutes  268803.0        103531.0
k_Bacteria|p_Firmicutes|c_CFGB3005  93723.0  0.0
k_Bacteria|p_Actinobacteria|c_Coriobacteriia  66279.0  103313.0
k_Bacteria|p_Firmicutes|c_Erysipelotrichia  43992.0  53855.0
k_Bacteria|p_Actinobacteria|c_Actinomycetia  8033.0  2613.0
k_Bacteria|p_Proteobacteria|c_Gammaproteobacteria  14952.0  34954.0
k_Bacteria|p_Firmicutes|c_CFGB1292  4055.0  3786.0
k_Bacteria|p_Firmicutes|c_CFGB74494  402.0  0.0
k_Bacteria|p_Firmicutes|c_Clostridia|o_Eubacteriales  8734167.0        7463924.0
k_Bacteria|p_Bacteroidetes|c_Bacteroidia|o_Bacteroidales  11205265.0        10998269.0
```

Desired output: one big table

HUMANN: FUNCTIONAL ANNOTATION

<https://github.com/biobakery/biobakery/wiki/humann3>

```
for file in sequences/concatenated/*.fastq.gz; do
  sample=$(basename "$file" ".fastq.gz")
  humann \
    --input sequences/concatenated/"${sample}.fastq.gz" \
    --output humann_out \
    --output-basename "${sample}" \
    --nucleotide-database $SCRATCH/humann_databases/chocophlan \
    --protein-database $SCRATCH/humann_databases/uniref \
    --metaphlan-options="-t=rel_ab --
index=mpa_vOct22_CHOCOPhlanSGB_202403 --
bowtie2db=$SCRATCH/metaphlan_databases" \
    --memory-use maximum \
    --threads=37 \
    --verbose
done
```

Very similar to MetaPhlAn

Default functional outputs:

- Proteins
- Pathways
- KO terms



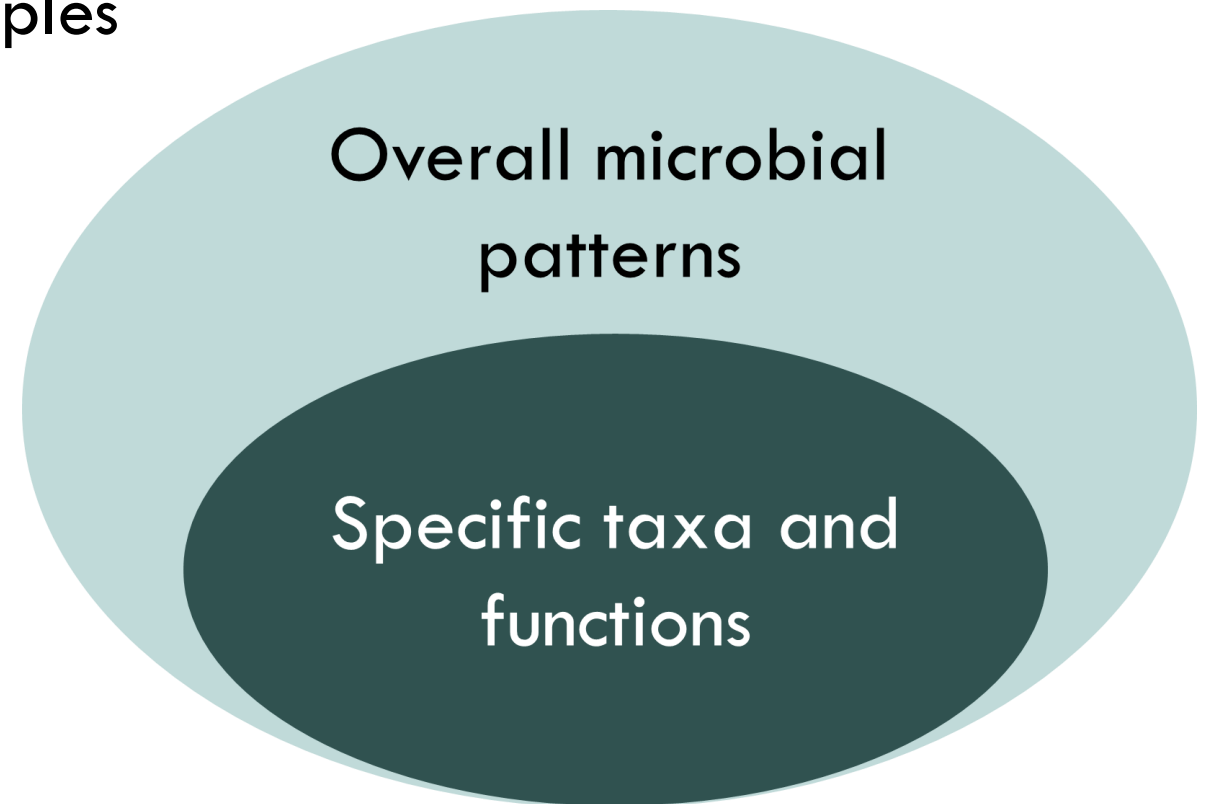
Note that it only allows relative abundance as an output

EXTRA NOTES ABOUT HUMANN

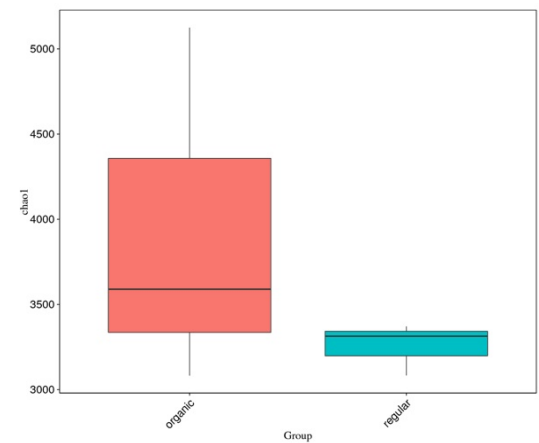
- Large number of outputs – consult manual for full explanations
- To speed up HUMAnN, MetaPhlAn outputs can be provided as inputs
 - Other similar tips & tricks available in the documentation
- Requires majority of the computing resources of the entire pipeline
 - Refine your approach where possible

TOPICS COVERED TODAY

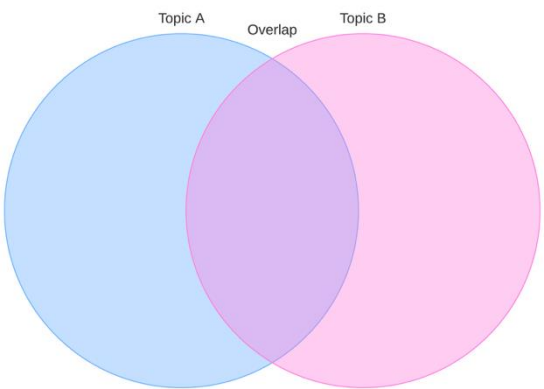
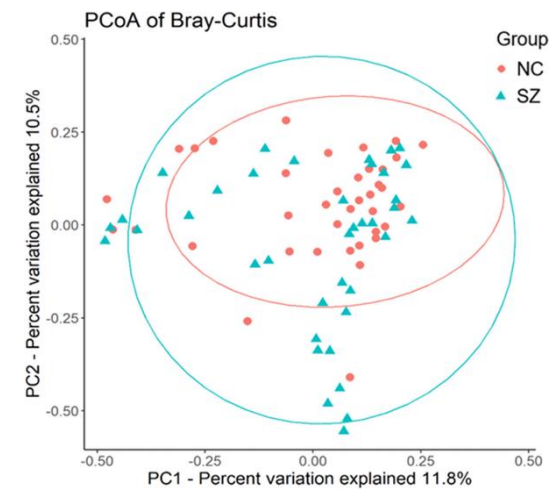
1. Metagenomic sequencing principles
2. Raw Data Processing (Tutorial)
3. **Key R Analyses (Tutorial)**
 1. Alpha and beta diversity
 2. Taxonomy bar plots
 3. Core microbiome
 4. Differential abundance
 5. Miscellaneous tools



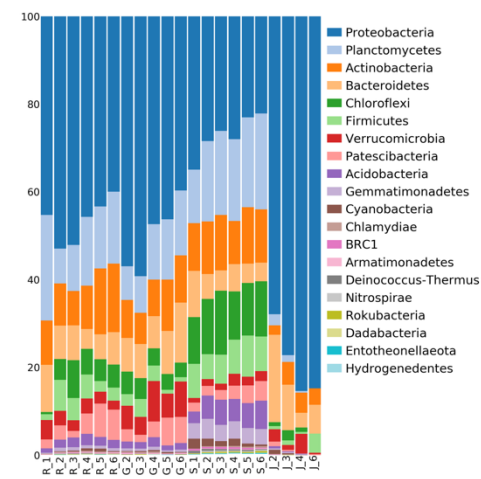
ALPHA DIVERISTY METRIC



BETA DIVERISTY METRIC



CORE MICROBIOME



TAXONOMY BAR PLOT

OVERALL MICROBIAL PATTERNS: DIVERSITY METRICS

All diversity metrics use **rarefied data**.

Alpha diversity: how diverse is each individual sample?

- Richness: # of features
- Evenness: are the features present in even proportions?

Beta diversity: how similar is each sample to the others?

- Compares microbial 'fingerprint'



Rich and even



Rich, not even



Not rich or even



Loop

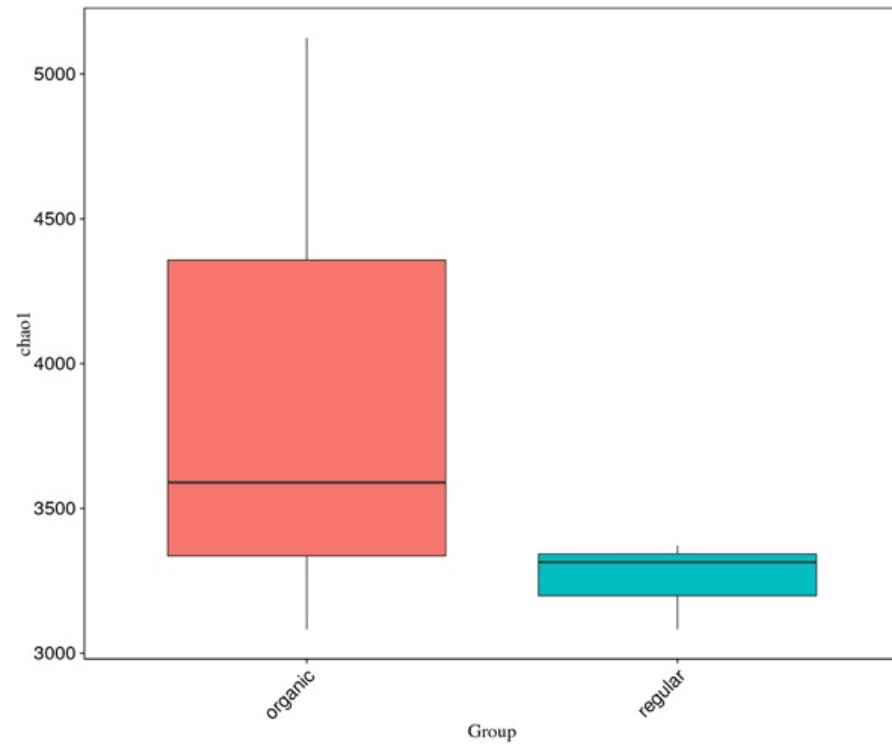


Whorl

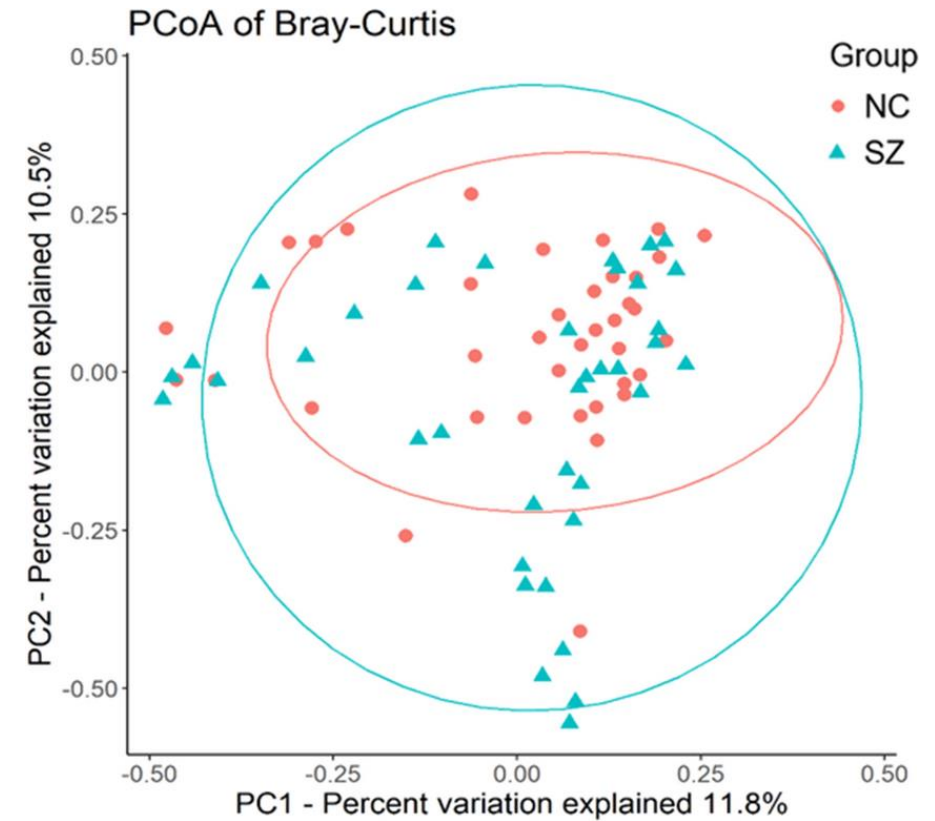


Arch

ALPHA DIVERISTY METRIC



BETA DIVERISTY METRIC



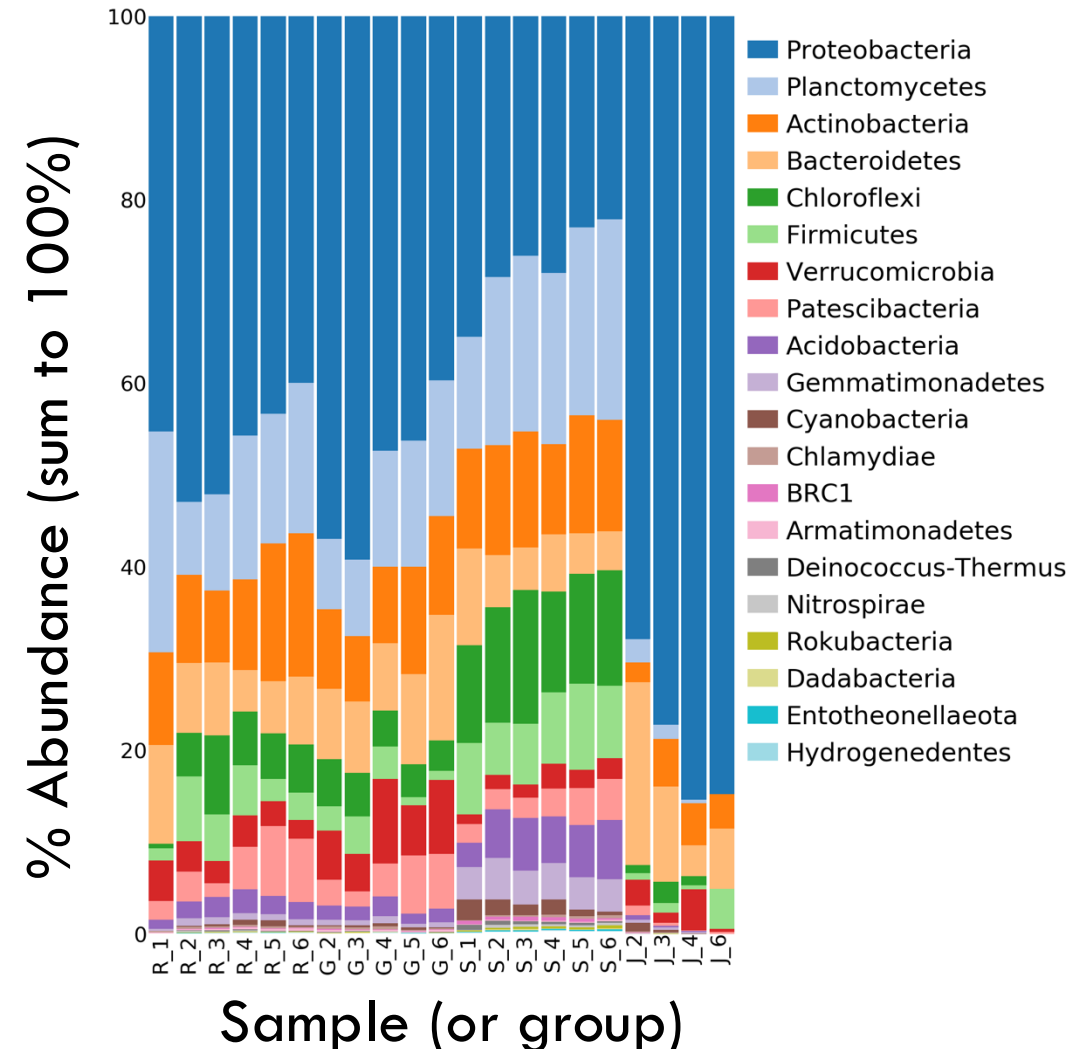
OVERALL MICROBIAL PATTERNS: TAXONOMY BAR PLOTS

Purpose:

- First-pass overview of microbial trends
 - **NOT a statistical approach**
 - Alpha diversity: visualizes evenness (& richness)
 - Beta diversity: visualizes similar trends
- Not commonly used for pathways
 - Too many pathways to plot

Plot characteristics:

- Taxa <1% often combined
- Abundances are stacked (add up to 100%)
- Individual samples best, groups OK

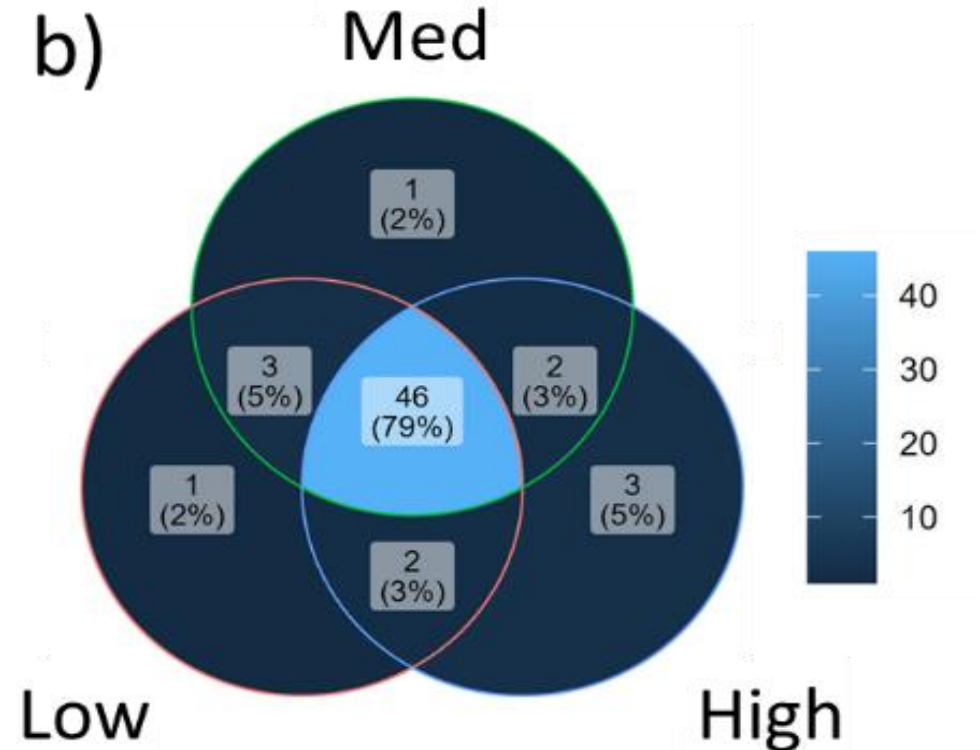


OVERALL MICROBIAL PATTERNS: CORE MICROBIOME

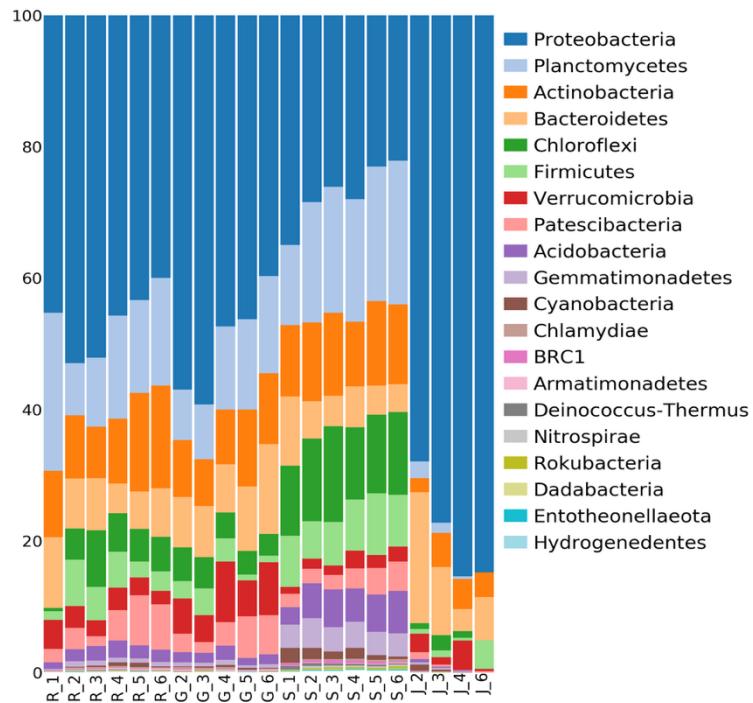
- Venn diagram of taxa or functions
- One circle per group
- Rarefied data used as input
 - Abundance and prevalence filters often applied (no standard cutoffs)

Gives information on:

- Alpha diversity: richness (# features)
- Beta diversity: proportion shared features



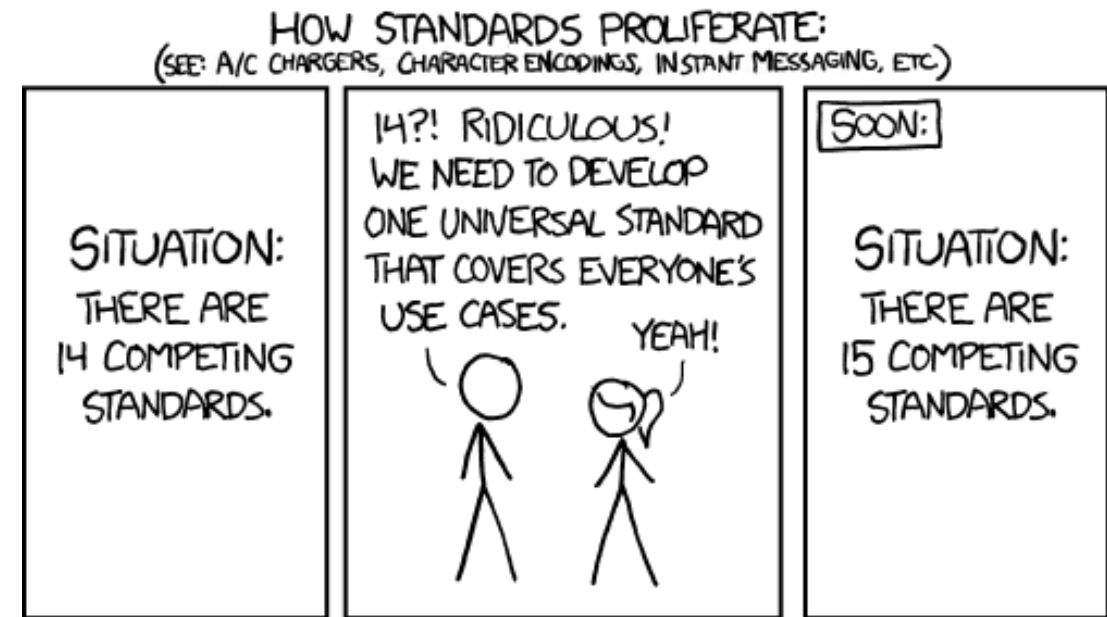
SPECIFIC DIFFERENCES: DIFFERENTIAL ABUNDANCE (DA) TOOLS



Statistically test each feature
(taxon or function)

SPECIFIC DIFFERENCES: DIFFERENTIAL ABUNDANCE (DA) TOOLS

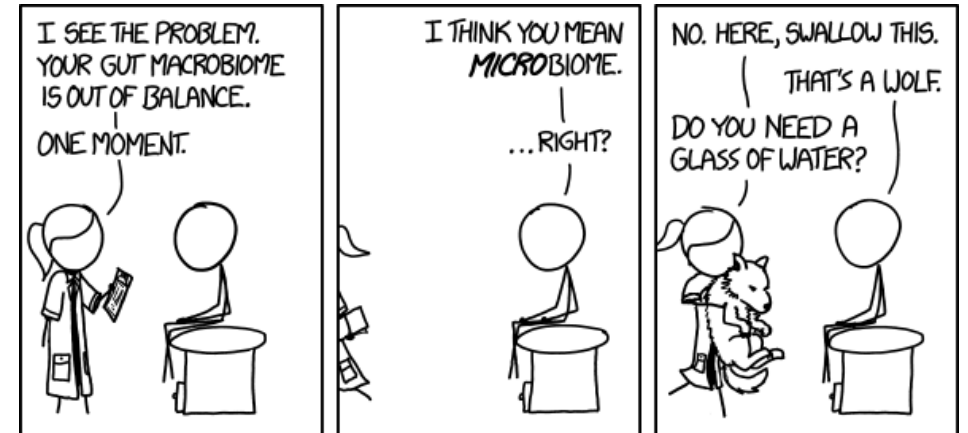
- Can include covariables
 - Some tools allow mixed modeling (ex. time series)
- Each tool addresses some (not all) of the statistical limitations of sequencing data
 - Overabundance of zeros, compositional abundances, etc
- Certain tools work better than others
 - <https://www.nature.com/articles/s41467-022-28034-z>
- Each tool has unique benefits/drawbacks
 - Features that are significant across multiple tools are more likely to be true positives



Relevant XKCD

GENERAL WORKFLOW FOR DIFFERENTIAL ABUNDANCE

1. Choose DA tool(s) and appropriate parameters/requirements:
 1. Data format/normalization/transformation
 2. Should abundance/prevalence filters be applied?
2. Run univariate and/or multivariate models
3. If applicable, compare results of multiple DA tools
4. Plot results
 - Relative abundance data are very commonly used for plots



Relevant XKCD

LOCOM

MaAsLin2

ALDEx2

EdgeR

DESeq2

LinDA

LEfSe

metagenomeSeq

ANCOM-BC

DA TOOL RECOMMENDATIONS

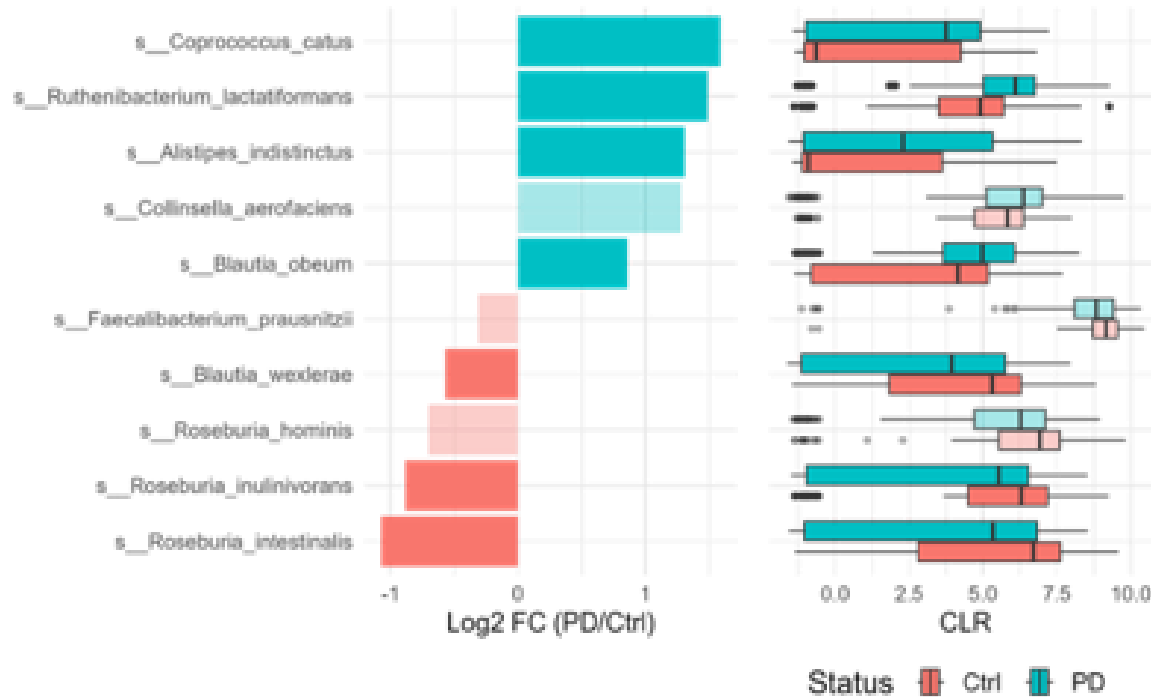
Combining results from 2 or 3 tools is recommended to ensure robustness!

- **ANCOM-BC**: good sensitivity and specificity. Not effective when # features ≤ 5 (ex. phylum level) or when dramatic microbial shifts are observed.
- **MaAsLin2**: linear modeling approach. Overestimates significance, but works with relative abundance data and is most used tool for testing functional data.
- **ALDEx2**: underestimates significance, but hits usually match very well with other tools. Powerful but very slow. Common to get few/no hits. Only works with count data (pseudo-count data needs to be rounded).
- **LinDA**: similar approach to ANCOM-BC and MaAsLin2. New-ish, performs very well but hasn't been formally compared to other tools.
- **DESeq2**: *Very commonly used, but is designed for RNAseq and is outdated; slowly getting phased out of DNA analysis*

PLOTTING RESULTS

Typical approach:

- DA tool(s) are used to calculate **significance**
 - Must** use adjusted P values, not raw P values
- Counts are transformed to **relative abundance** or **CLR** (see figures)
 - Significant features are plotted as **boxplots** or as **log 2 fold changes**
 - Often then correlated with other metadata using **standard statistical approaches**
 - ex. linear models, Wilcoxon - arguably not ideal, but definitely common



OTHER COMMON MICROBIOME ANALYSES

Indicator taxa: features that are not just higher, but highly SPECIFIC to, a given group

Random forest: can some combination of features predict an outcome variable?

Networks/correlation matrices: What features are interconnected, how much 'communication' there is between features in the dataset

Stratified functional analysis: what taxa does this function belong to?

